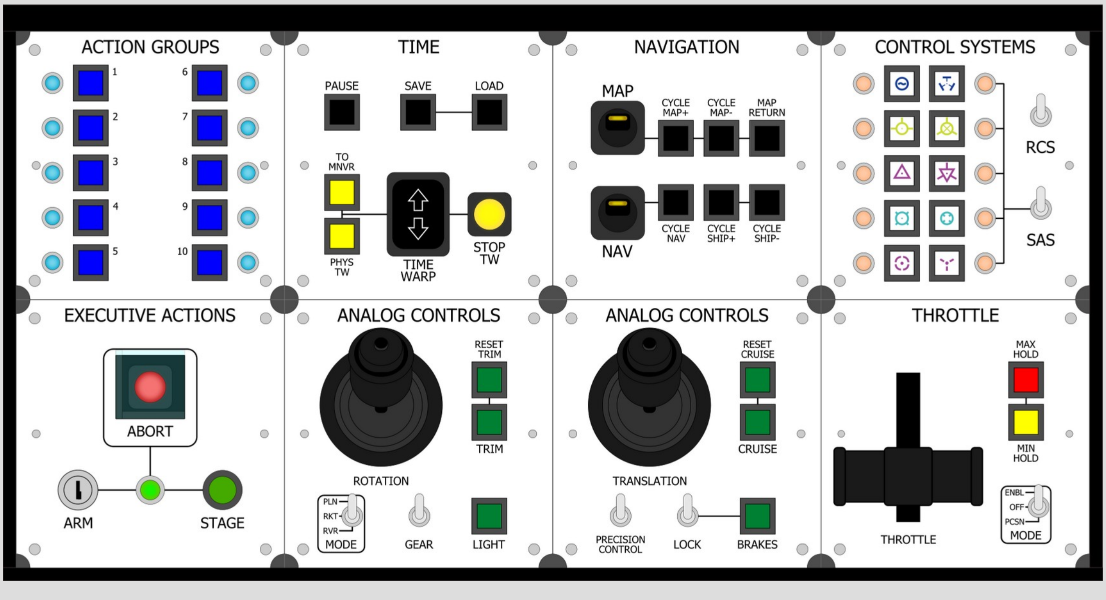
Untitled Space Craft

Complete Documentation



Bill of Materials

All parts are through-hole, 2.54mm pitch unless otherwise stated.

The links provided are not necessarily accurate. It is up to you to read datasheets to make sure the part will suit your needs. Common pitfalls include switches that latch rather than momentary, wrong hole dimensions, etc.

Module	Part	Qty	Value	Note
All Modules	2.0mm Pitch 16pin Connector	1	Socket	Standard module connector
	Electrolytic Cap	1	10uf 16v	Main power decoupling
	Round Neodymium Magnets	4	5mm/5mm	Requires high precision
	PCB	1		
	Acrylic Face Plate	1		
	Screw	2	3M, 15mm	Telemetry needs 4
	Nut	4	3M	Telemetry needs 16
Abort	Circle Button	1	Red, 12mm	OFF-(ON)
	Safety Switch	1	Red	
	LED	1	Red, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Action Groups	Square Button	10	Blue	
	LED	10	Blue, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	10k, 1/4W	
	Resistor	10	62k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
	74HC595	2		
Analog	Square Button	2	Blue	
	Switch	2	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	7	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
Analog Throttle	Square Button	2	Blue	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
	Potentiometer	1	B20K	
	Potentiometer Cover Dial	1		
Camera	Square Button	4	Black	
	Square Button	6	White	
	Chip Socket	1	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	10	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	1		
	74HC165	2		

Module	Part	Qty	Value	Note
Control Systems	Square Button	10	White	
	Switch	2	SPST	
	LED	10	Orange, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	22	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
	74HC165	2		
Executive Actions (Control)	Square Button	1	White	
	Square Button	2	Blue	
	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Switch	2	SPST	
	Key Switch	1	12mm	
	LED	3	Orange, 3mm	
	LED	2	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	10	10k, 1/4W	
	Resistor	2	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions (Groups)	Square Button	5	Blue	
	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Key Switch	1	12mm	
	LED	5	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	7	10k, 1/4W	
	Resistor	5	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	24x26mm	
	Key Switch	1	12mm	
	LED	1	Green, 5mm	
	LED Socket	1	5mm	
	Chip Socket	1	8 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.2K, 1/4W	
	Resistor	2	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		

Module	Part	Qty	Value	Note
EVA	Square Button	6	White	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	11	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC1595	2		
	74HC165	2		
Navigation	Square Button	6	Black	
	LED Switch	2	Yellow	
	Chip Socket	1	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	2	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny45	1		
	74HC165	1		
Natigation (Time)	Square Button	2	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	D36 LED Switch	1	Yellow	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Rotation	Square Button	3	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
Rotation (Throttle)	Square Button	2	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	4	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	Potentiometer	1	B20K	
	Potentiometer Cover Dial	1		

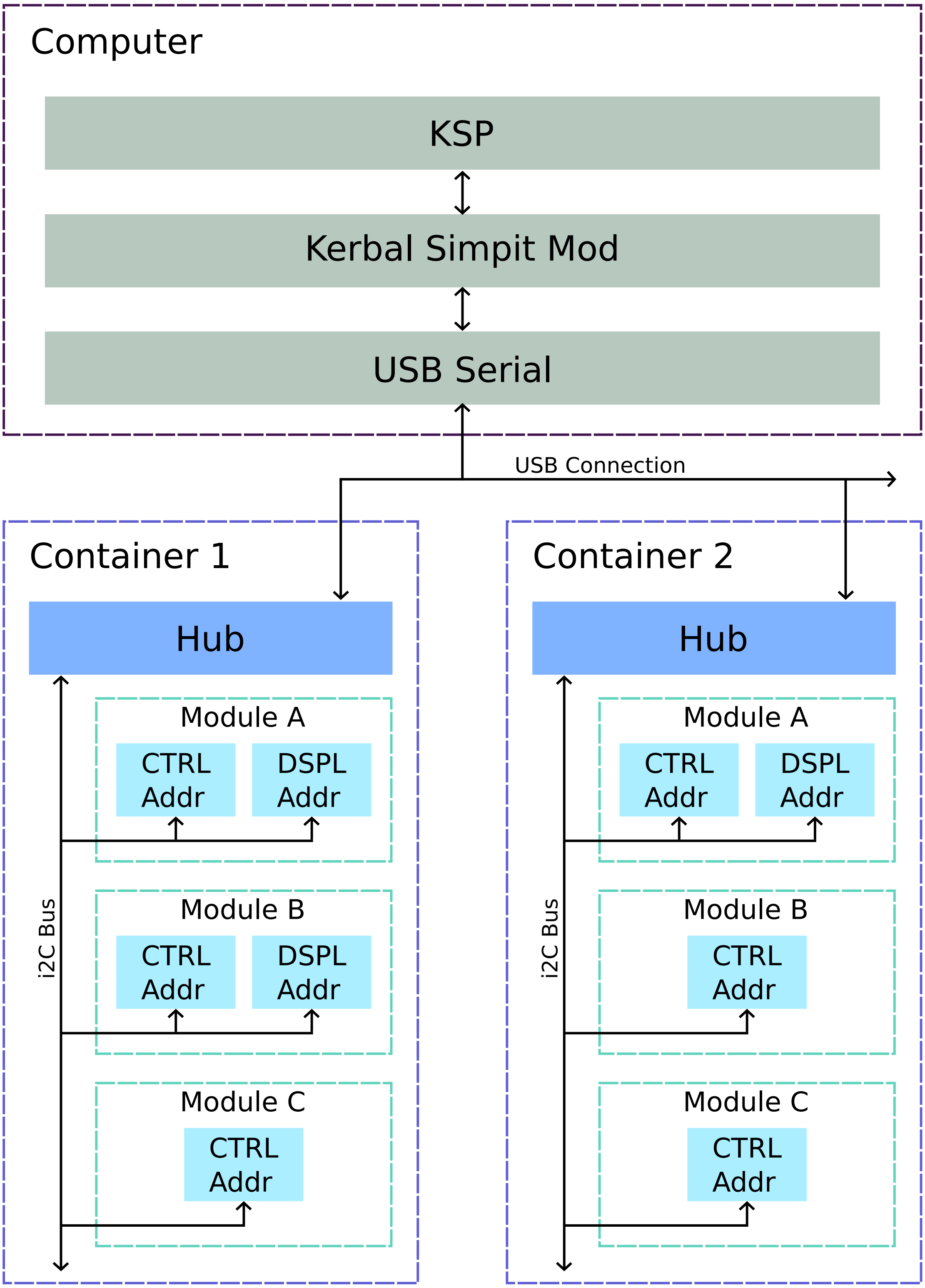
Module	Part	Qty	Value	Note
Stage	Circle Button	1	Green, 12mm	OFF-(ON)
	Safety Switch	1	Green,	
	LED	1	Green, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	150, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Telemetry	Square Button	4	Black	
	LCD Screen	1	128x64	ST7920 driven
	Header Pins	1	1x20 Pin	Not socket
	Resistor	4	10k, 1/4W	
	Washer	8	3M	Depends on thickness
	Arduino Nano	1		
Throttle	Square Button	1	Yellow	
	Square Button	1	Red	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	Potentiometer	1	B20k, 6 Pin	Two parallel channels
	Throttle Lever	1		3D printed
	Throttle Friction Mechanism	1		3D printed
	Chip Socket	1	8 Pin	
	Resistor	2	10k, 1/4W	
	Diode	1	1N4148	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Time	Square Button	3	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Translation	Square Button	3	Green,	
	Switch	2	SPST	
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Resistor	5	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny24	1		
Utility (Navigation)	Square Button	2	Black	
	Square Button	5	Green	
	D36 LED Switch	1	Yellow	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		

Module	Part	Qty	Value	Note
Utility (Time)	Square Button	3	Black	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		
Quad Hub	2.0mm Pitch 16pin Connector	4	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Octo Hub	2.0mm Pitch 16pin Connector	8	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Miscellaneous	Power Supply	1	5V	Center 5V, Sleeve ground
	Micro USB cable	1		
	Neodymium Magnets	10	5mm/5mm	
	Ribbon Cable	2-3m	16 wire, 1mm	Depending on size
	Ink	1	Black	Alcohol or acetone based
	Acrylic Container	1		Depending on size
	2.0mm Pitch 16pin Connector	2-16	Pin	Two per module

System Diagram

Untitled Space Craft utilizes the Kerbal Simpit Revamped mod for KSP in order to communicate with the game. The mod talks to controllers over the serial bus. Multiple containers can be connected to Simpit by editing the settings.cfg file in the GameData/KerbalSimpit folder.

Each container has a Hub, powered by Arduino Micro, with up to 8 modules driven by ATtiny microcontrollers. They communicate via the i2C protocol. Each module has one (for *just control*) or two (*for control and display*) addresses, designated with **CTRL** and **DSPL**.



I2C Addresses

Module	Address
Special	0-7
	8
	9
Stage	10
Abort	11
Executive Actions	12
Camera	13
Navigation	14
Navigation (Time)	15
Executive Actions (Groups) – Ctrl	16
Executive Actions (Groups) – Dspl	17
Executive Actions (Control) – Ctrl	18
Executive Actions (Control) – Dspl	19
Time – Ctrl	20
Time – Dspl	21
EVA - Ctrl	22
EVA – Dspl	23
Control Systems – Ctrl	24
Control Systems – Dspl	25
Utility Navication	26
Utility Time	27
	28
	29
Action – Ctrl	30
Action – Dspl	31
Action 2 – Ctrl	32
Action 2 – Dspl	33
Action 3 – Ctrl	34
Action 3 – Dspl	35
	36
	37
	38
	39
Telemetry	40
	41
	42
	43
	44
Analog (Throttle)	45
Rotation (Throttle)	46
Translation	47
Analog	48
Rotation	49
Throttle	50
	51
	52
	53
	54
Fuel – Ctrl	55
Fuel – Dspl A	56
Fuel – Dspl B	57
	58
	59
Velocity – Ctrl	60
Velocity – Dspl	61
Radar – Ctrl	62
Radar – Dspl	63
Apsides – Ctrl	64
Apsides – Dspl	65

Module	Address
	66
	67
	68
	69
System – Ctrl	70
System – Dspl/Snd	71
	72
	73
	74
	75
	76
	77
	78
	79
	80
	81
	82
	83
	84
	85
	86
	87
	88
	89
Incompatible	90-119
Special	120-127

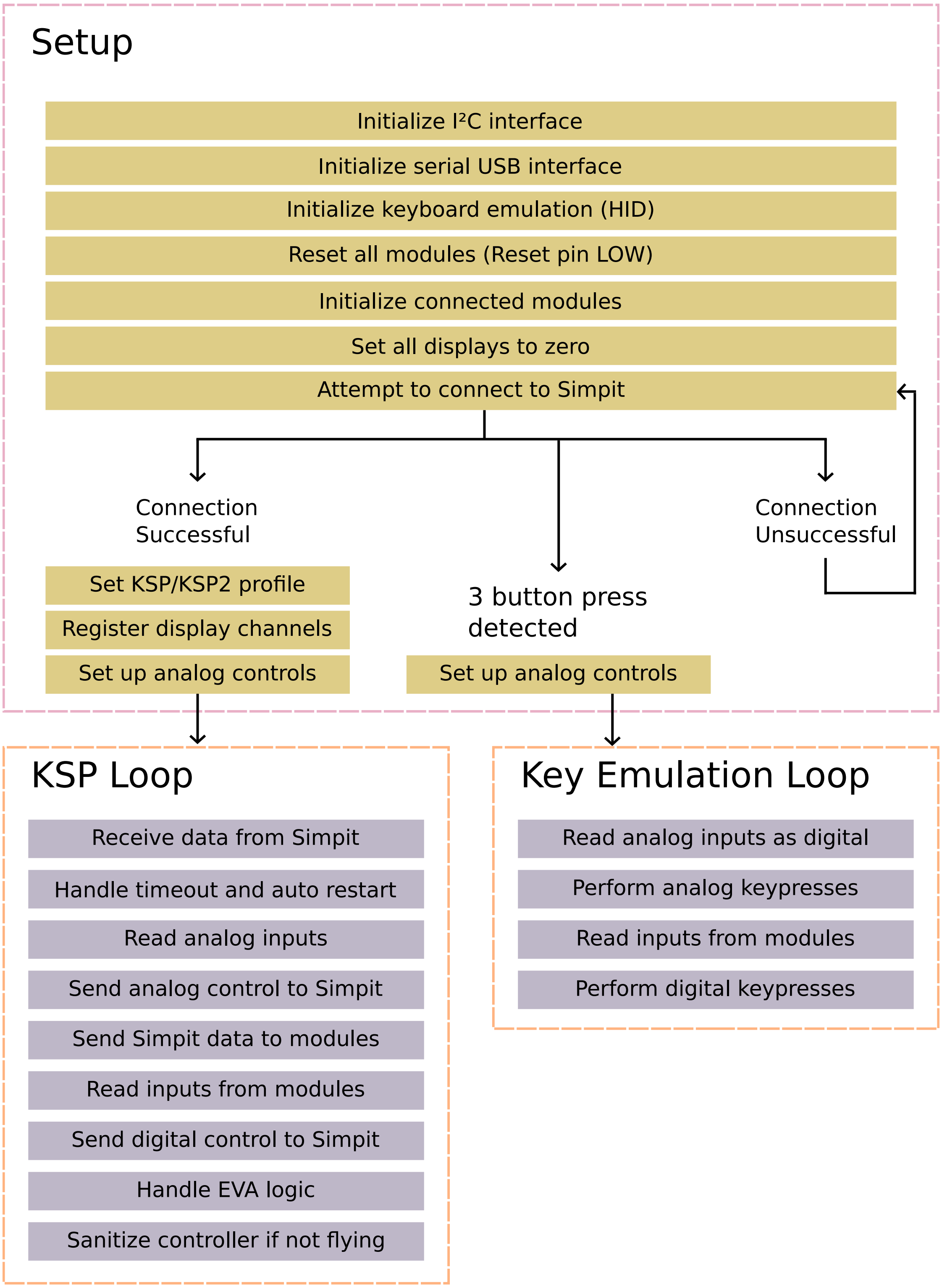
Purpose	Range
Input/Output	10 – 44
Analog	45 – 54
Display	55 – 89

Version
1.0
2.0
3.0
4.0
5.0
6.0

Firmware Execution Flow

Untitled Space Craft firmware is written in C/C++ and can be easily managed with the Arduino IDE. When the controller is powered on, the hub will perform a setup sequence where it manages libraries and initializes the modules.

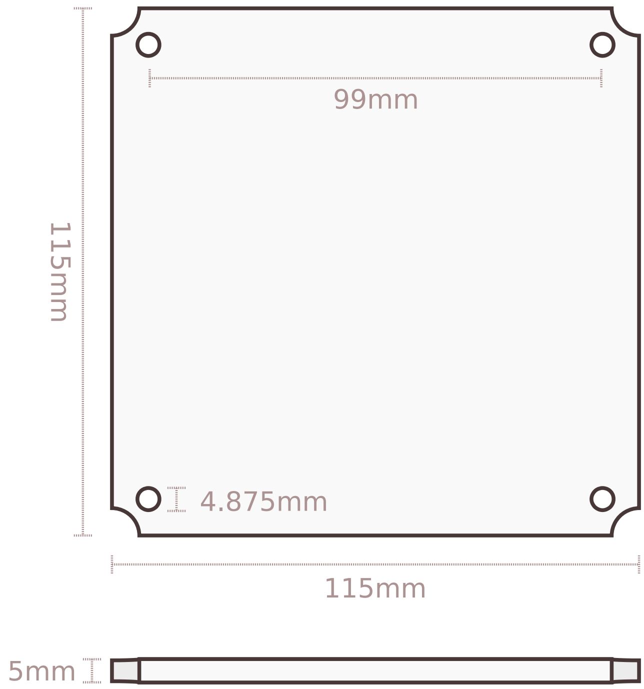
Once it is ready, it will repeatedly attempt to connect to Simpit for either KSP1 or KSP2. During this time, if 3 repeated inputs are detected, it will enter Keyboard Emulation mode. If the connection to Simpit is successful, it will enter KSP mode.



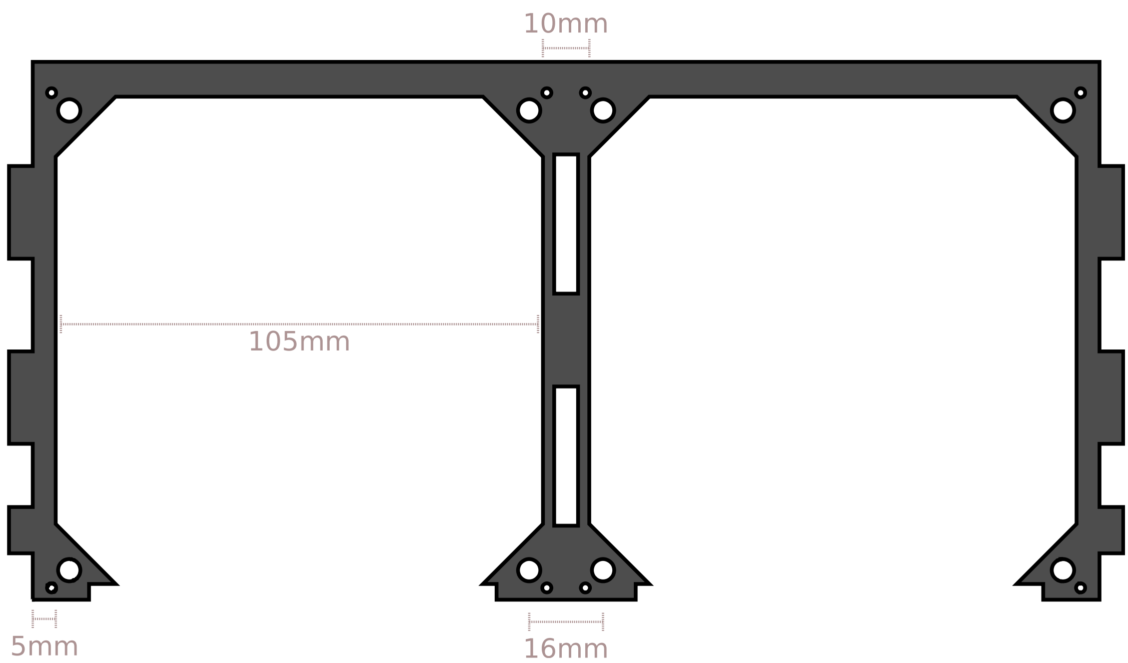
The Standard Module

The standard module is the form factor that governs the design of USC. Each module follows this same basic body plan.

It is 115mm square, with holes that are 4.875mm in diameter and 99mm apart from their centers, and 5mm thick. The hole diameter is a function of the diameter of the neodymium magnets, the flexibility of the medium (typically acrylic), and the thickness and settings of the laser. It is advised to do a test cut first to guarantee a proper friction fit.

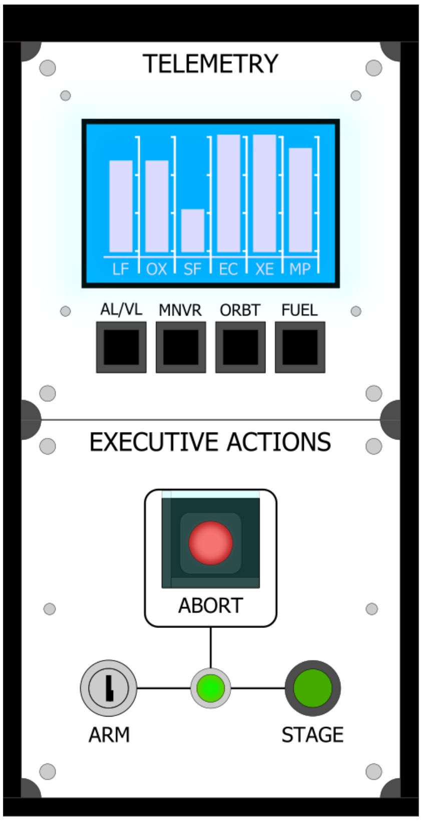


The module fits into the Module Slot. They have the same holes for magnets, 16mm apart where the modules touch. The slots are 105mm square, with a 5mm edge (or 10mm between them).

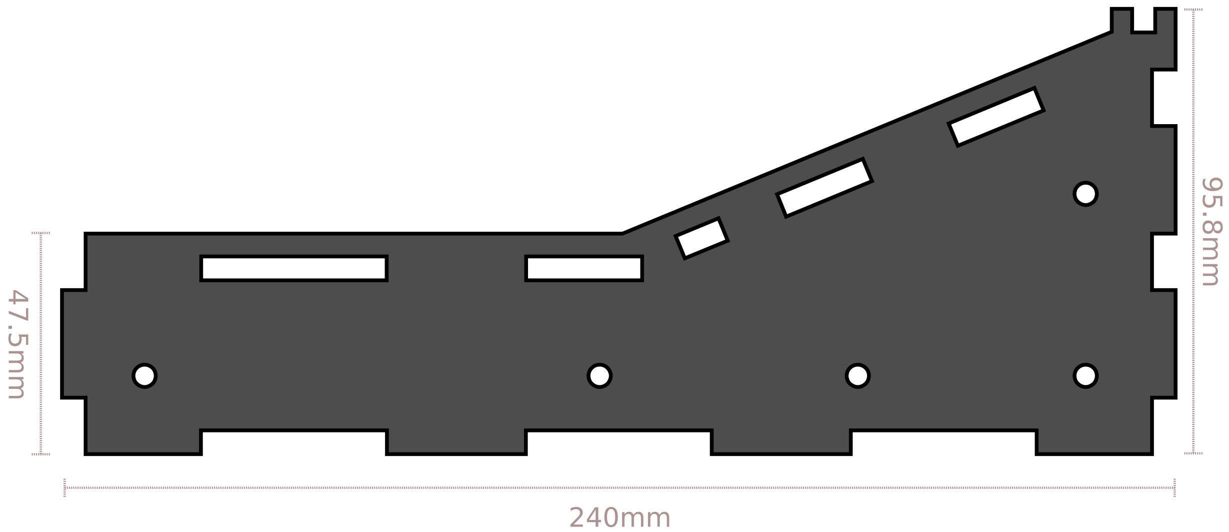


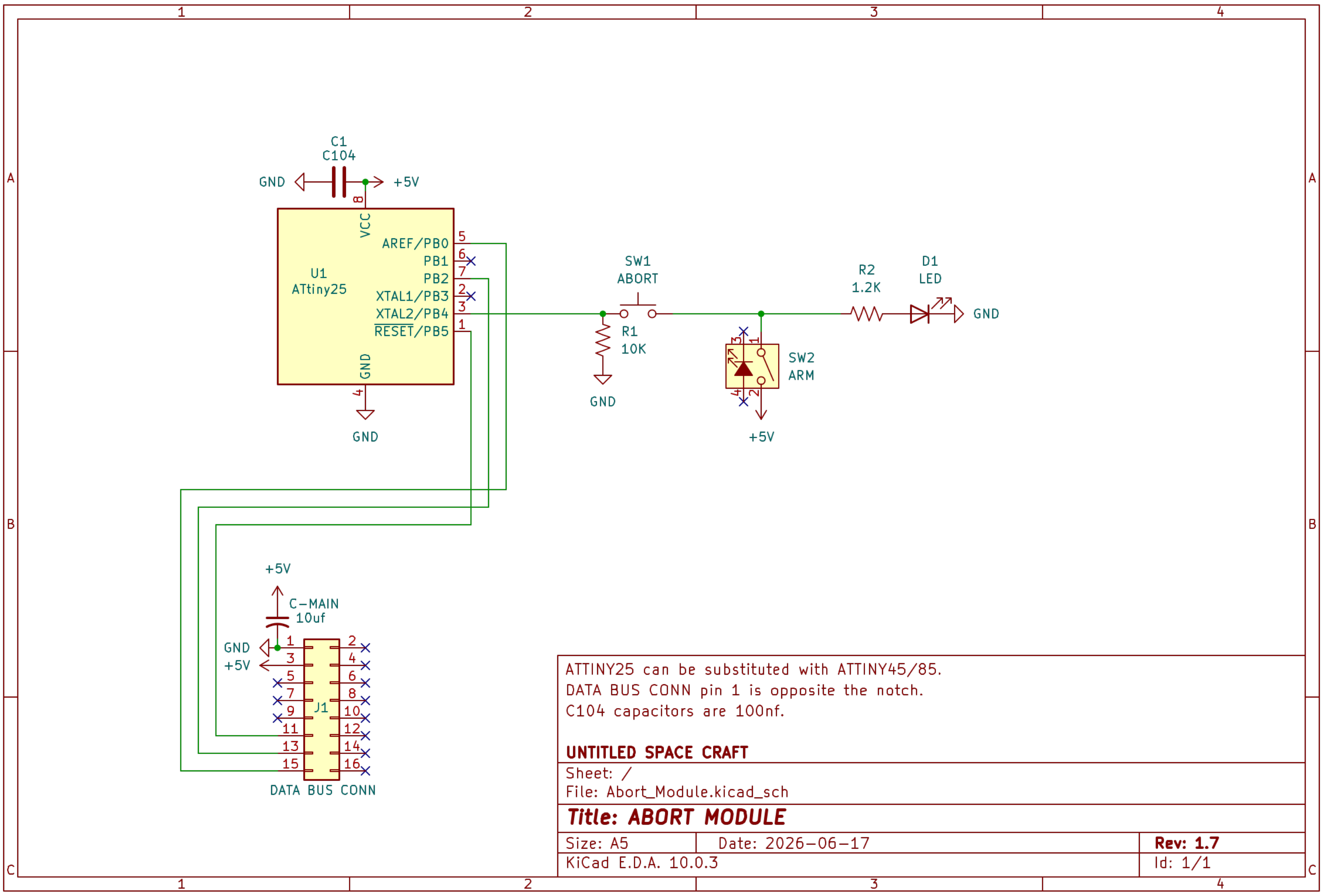
Containers

Containers are the larger units of USC. They may have up to 8 module slots and as few as 1.

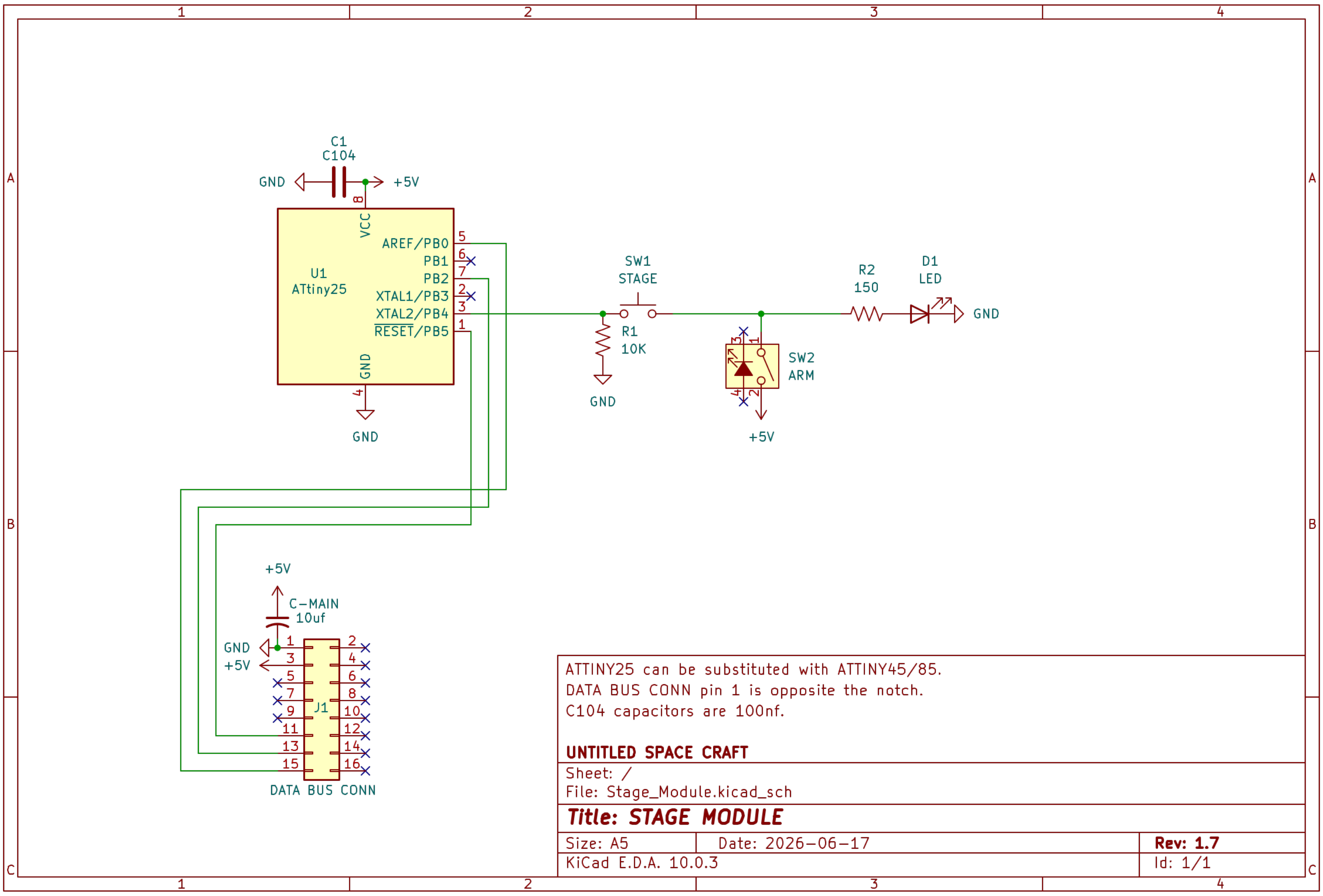


Containers have several measurements in common. **Their pieces should all be 5mm thick.** They are all 95.8mm tall on the back end. Containers that are two modules deep are 240mm long and 47.5mm tall on the short end. The measurements throughout the containers may not be exact; typically they are a few fractions of a millimeter shorter in order to account for the space needed for the caulk adhesive.

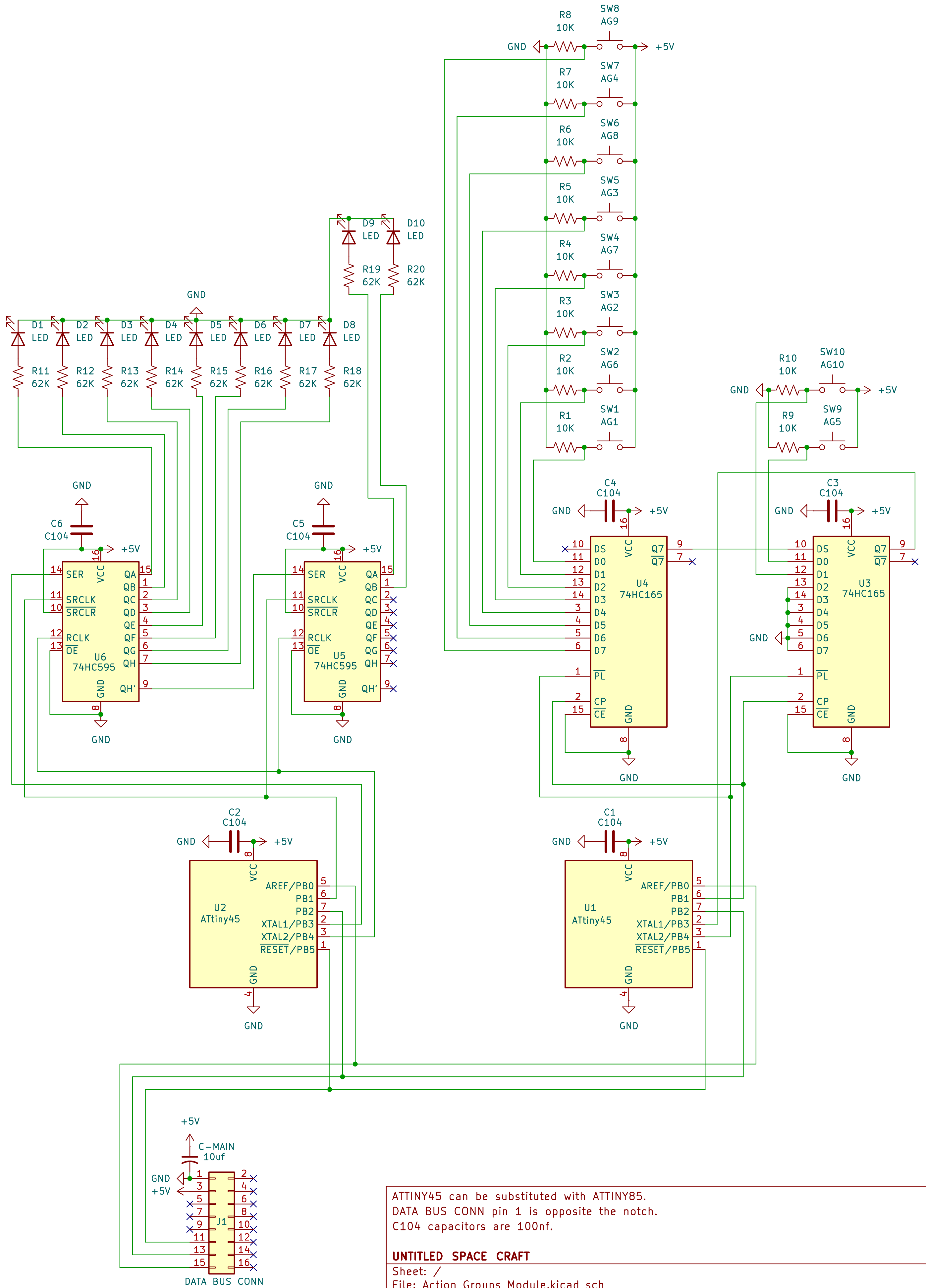




C1
C104
GND
+5V
VCC
U1
ATtiny25
AREF/PB0
PB1
PB2
XTAL1/PB3
XTAL2/PB4
RESET/PB5
GND
SW1
ABORT
R1
10K
R2
1.2K
D1
LED
SW2
ARM
C-MAIN
10uf
J1
DATA BUS CONN
ATTINY25 can be substituted with ATTINY45/85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.
UNTITLED SPACE CRAFT
Sheet: /
File: Abort_Module.kicad_sch
Title: ABORT MODULE
Size: A5
Date: 2026-06-17
Rev: 1.7
KiCad E.D.A. 10.0.3
Id: 1/1



C1
C104
GND
+5V
VCC
U1
ATtiny25
AREF/PB0
PB1
PB2
XTAL1/PB3
XTAL2/PB4
RESET/PB5
GND
SW1
STAGE
R1
10K
R2
150
D1
LED
SW2
ARM
C-MAIN
10uf
J1
DATA BUS CONN
ATTINY25 can be substituted with ATTINY45/85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.
UNTITLED SPACE CRAFT
Sheet: /
File: Stage_Module.kicad_sch
Title: STAGE MODULE
Size: A5
Date: 2026-06-17
Rev: 1.7
KiCad E.D.A. 10.0.3
Id: 1/1



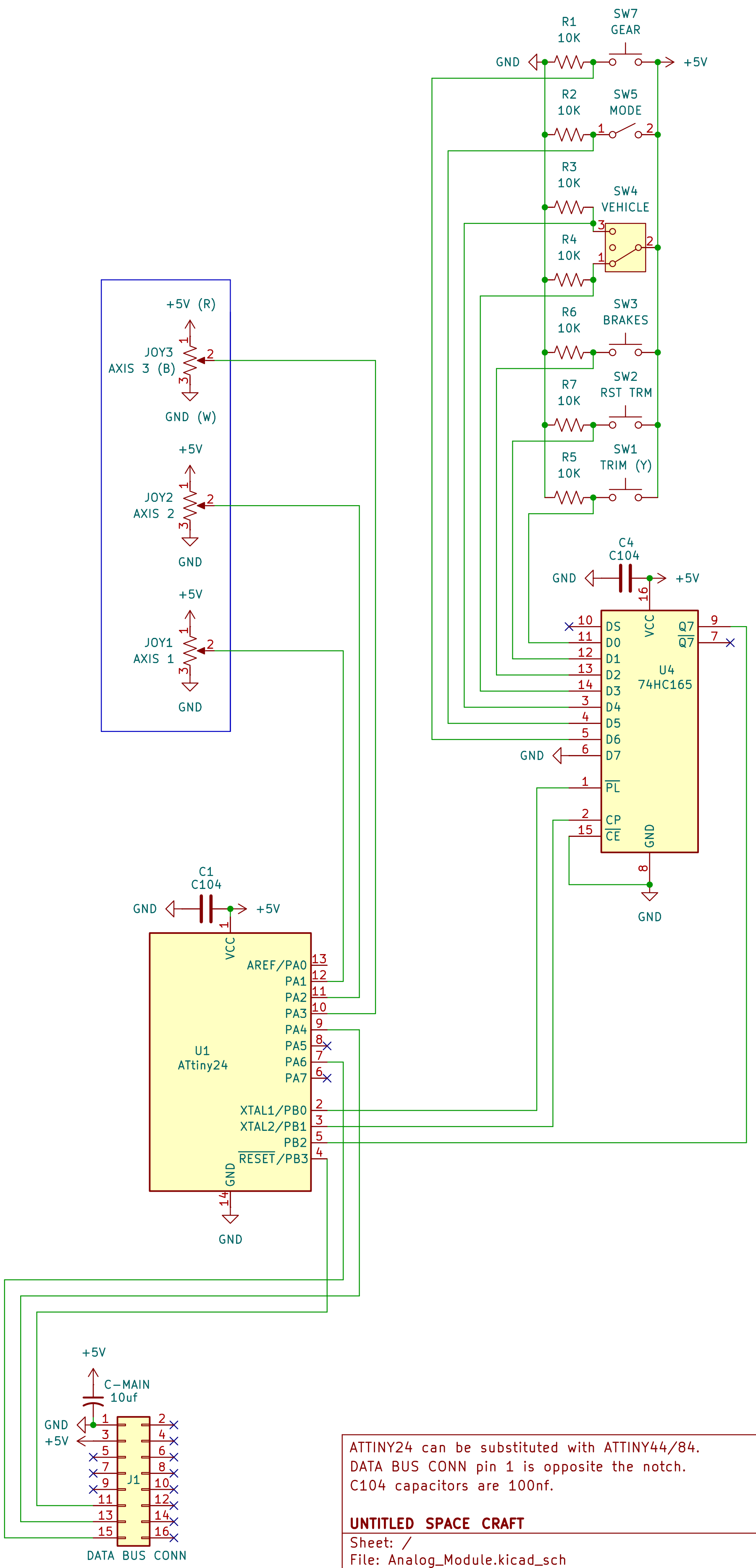
ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Action_Groups_Module.kicad_sch

Title: ACTION GROUPS MODULE

Size: A4 Date: 2026-06-17 **Rev: 1.7**
KiCad E.D.A. 10.0.3 Id: 1/1



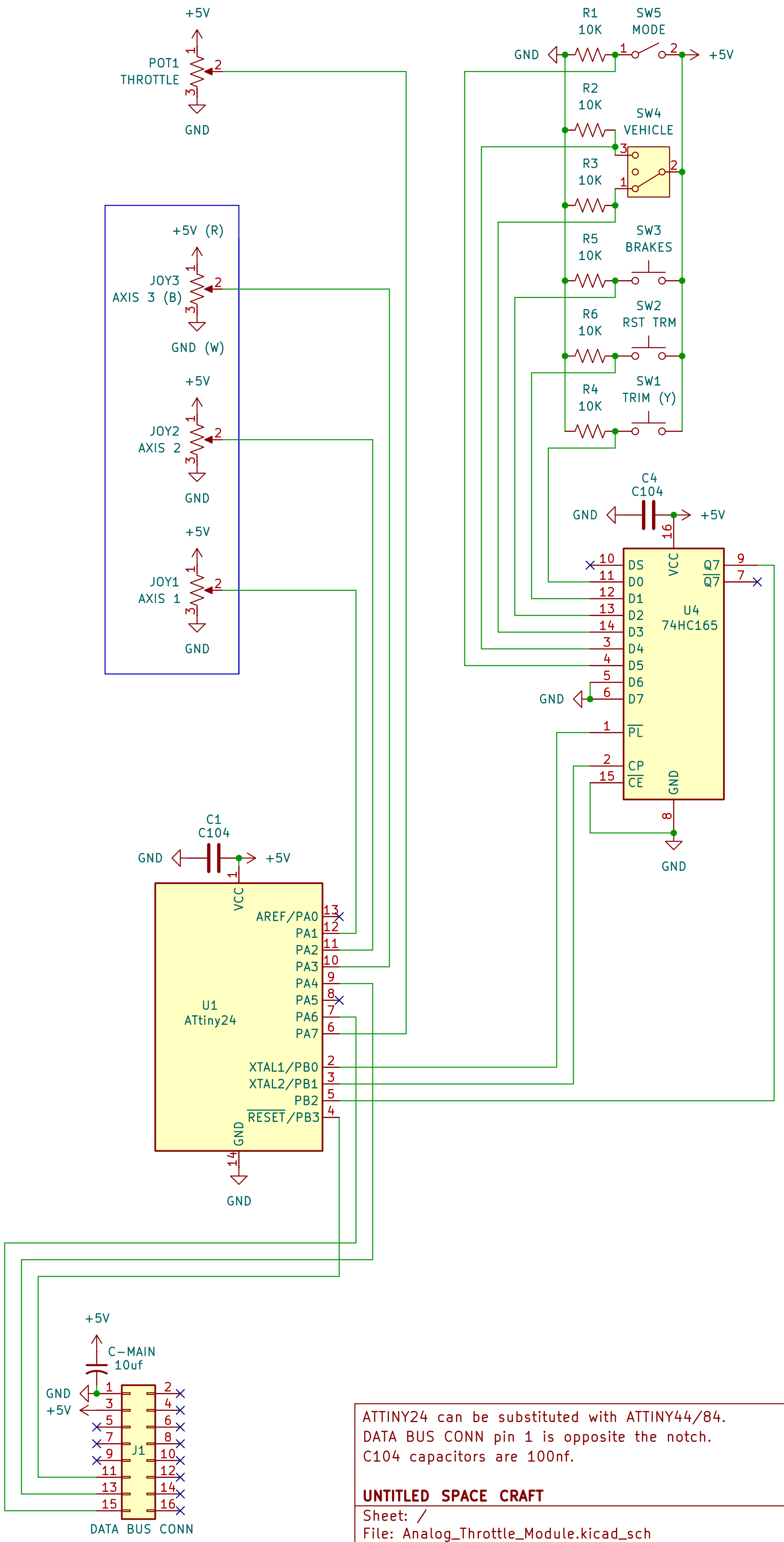
ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Analog_Module.kicad_sch

Title: ANALOG MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



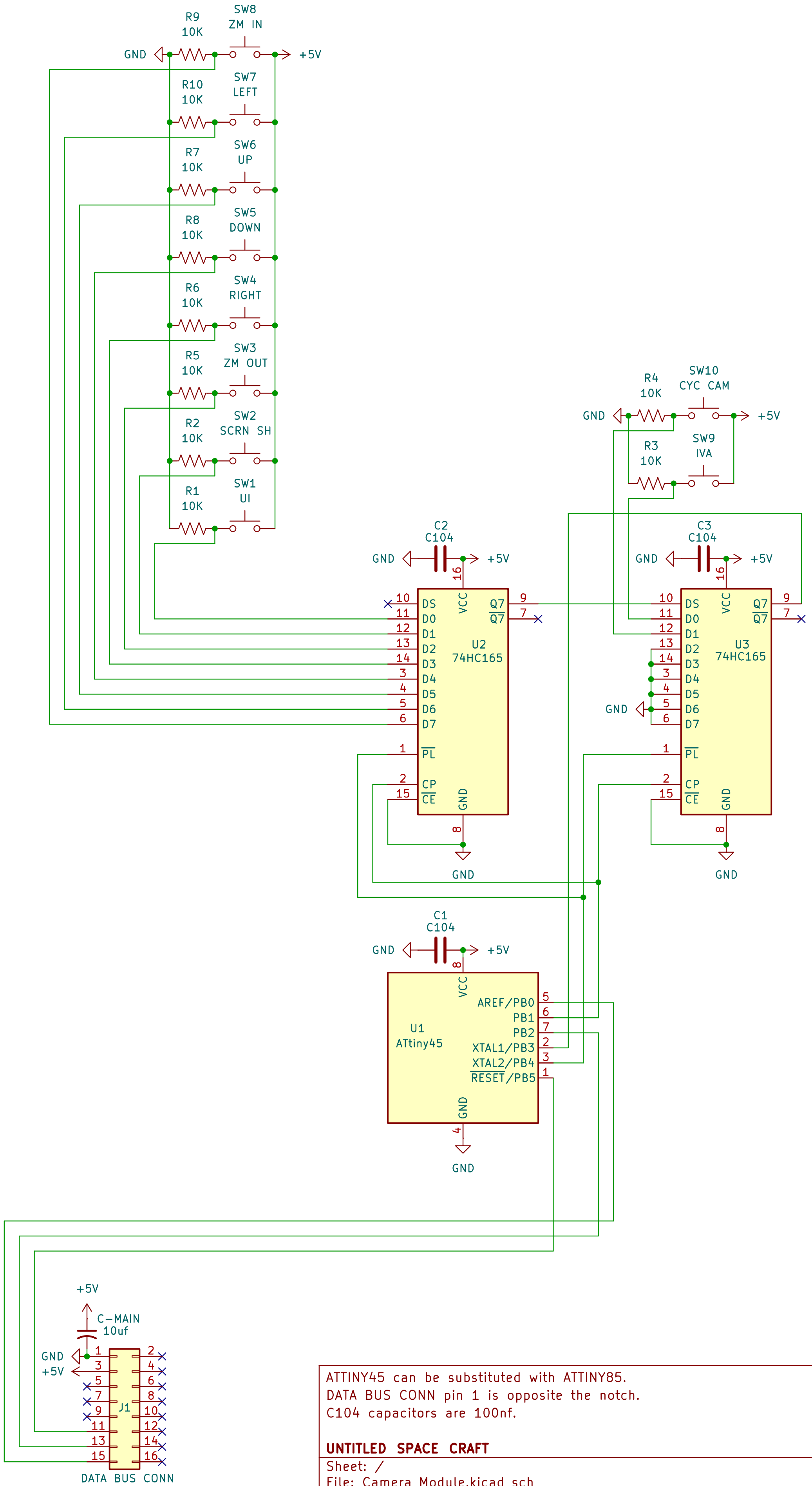
ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Analog_Throttle_Module.kicad_sch

Title: ANALOG THROTTLE MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



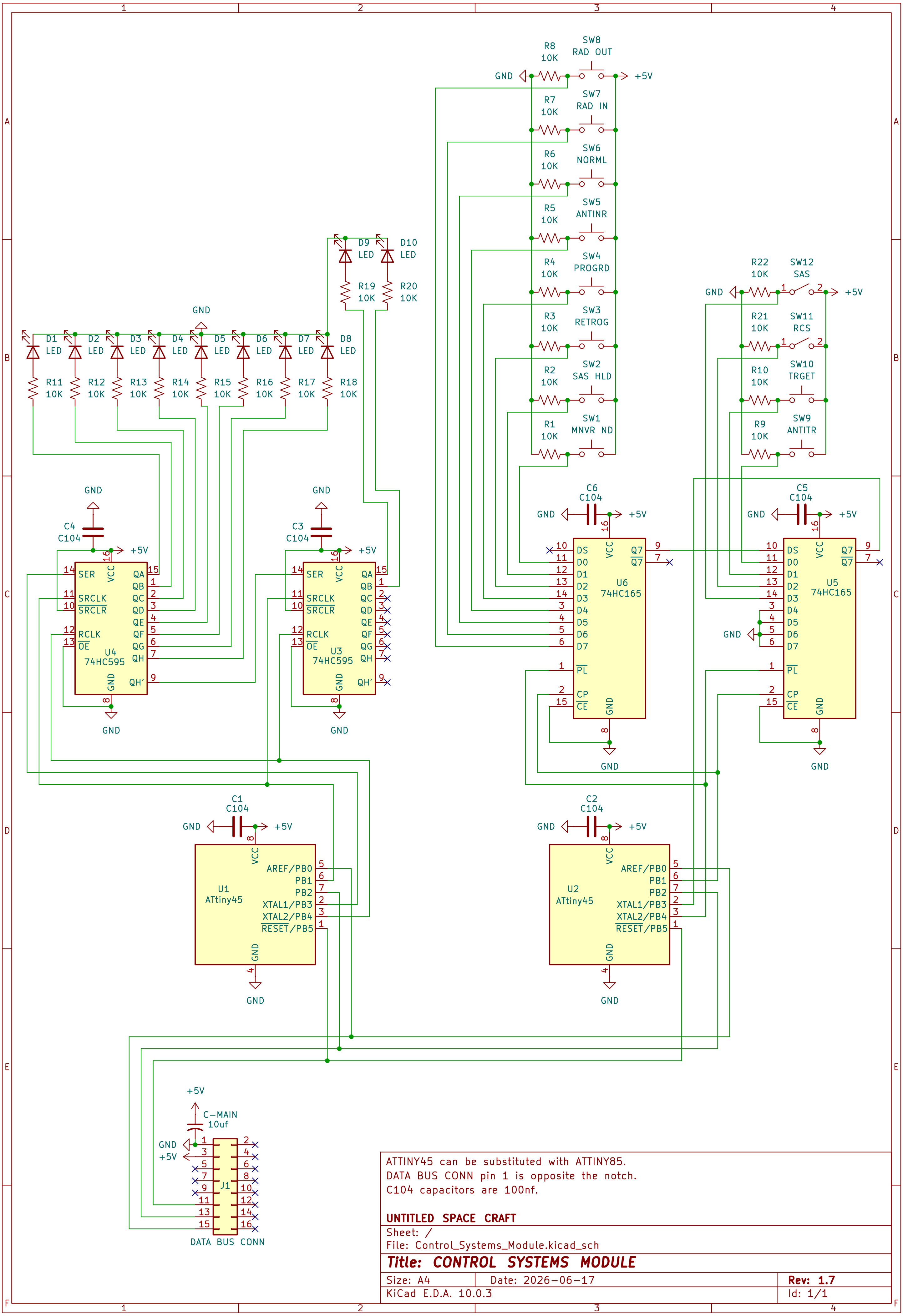
ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Camera_Module.kicad_sch

Title: CAMERA MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



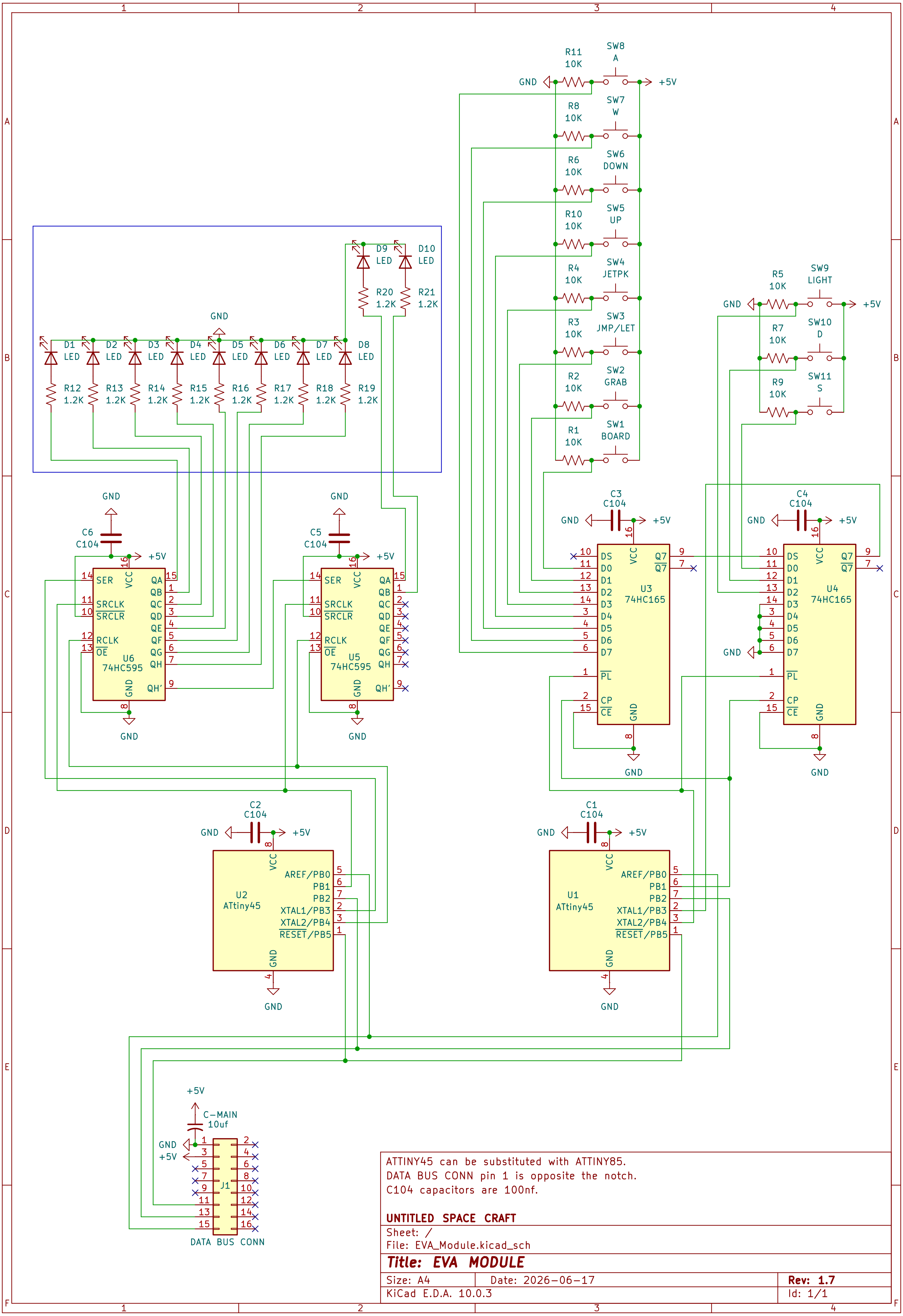
ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

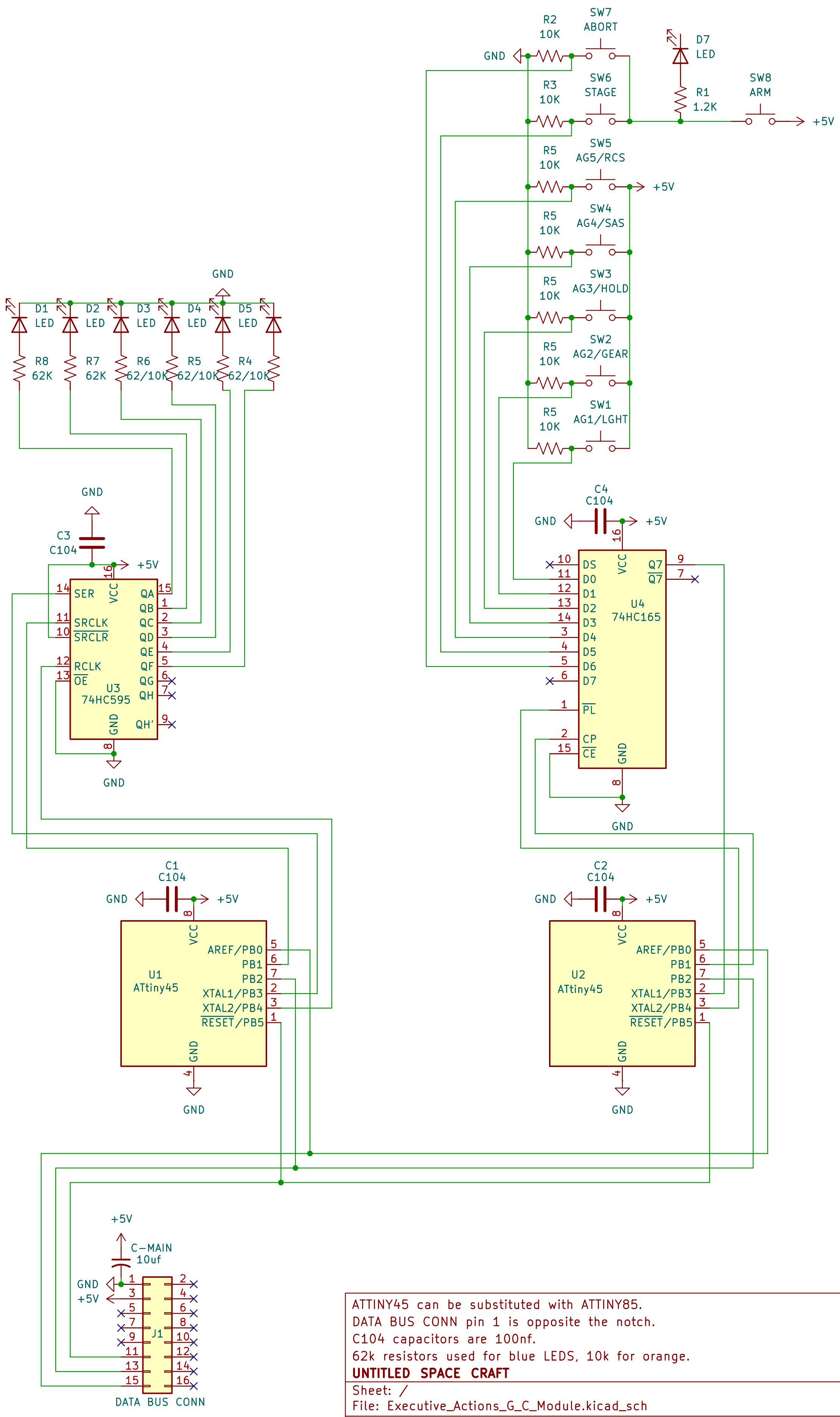
UNTITLED SPACE CRAFT

Sheet: /
File: Control_Systems_Module.kicad_sch

Title: CONTROL SYSTEMS MODULE

Size: A4 Date: 2026-06-17 **Rev: 1.7**
KiCad E.D.A. 10.0.3 Id: 1/1


SW8 A
SW7 W
SW6 DOWN
SW5 UP
SW4 JETPK
SW3 JMP/LET
SW2 GRAB
SW1 BOARD
R11 10K
R8 10K
R6 10K
R10 10K
R4 10K
R3 10K
R2 10K
R1 10K
SW9 LIGHT
SW10 D
SW11 S
R5 10K
R7 10K
R9 10K
D1 LED
D2 LED
D3 LED
D4 LED
D5 LED
D6 LED
D7 LED
D8 LED
D9 LED
D10 LED
R12 1.2K
R13 1.2K
R14 1.2K
R15 1.2K
R16 1.2K
R17 1.2K
R18 1.2K
R19 1.2K
R20 1.2K
R21 1.2K
C6 C104
C5 C104
C3 C104
C4 C104
C2 C104
C1 C104
U6 74HC595
U5 74HC595
U3 74HC165
U4 74HC165
U2 ATtiny45
U1 ATtiny45
C-MAIN 10uf
J1
DATA BUS CONN
GND
+5V
ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.
UNTITLED SPACE CRAFT
Sheet: /
File: EVA_Module.kicad_sch
Title: EVA MODULE
Size: A4
Date: 2026-06-17
Rev: 1.7
KiCad E.D.A. 10.0.3
Id: 1/1



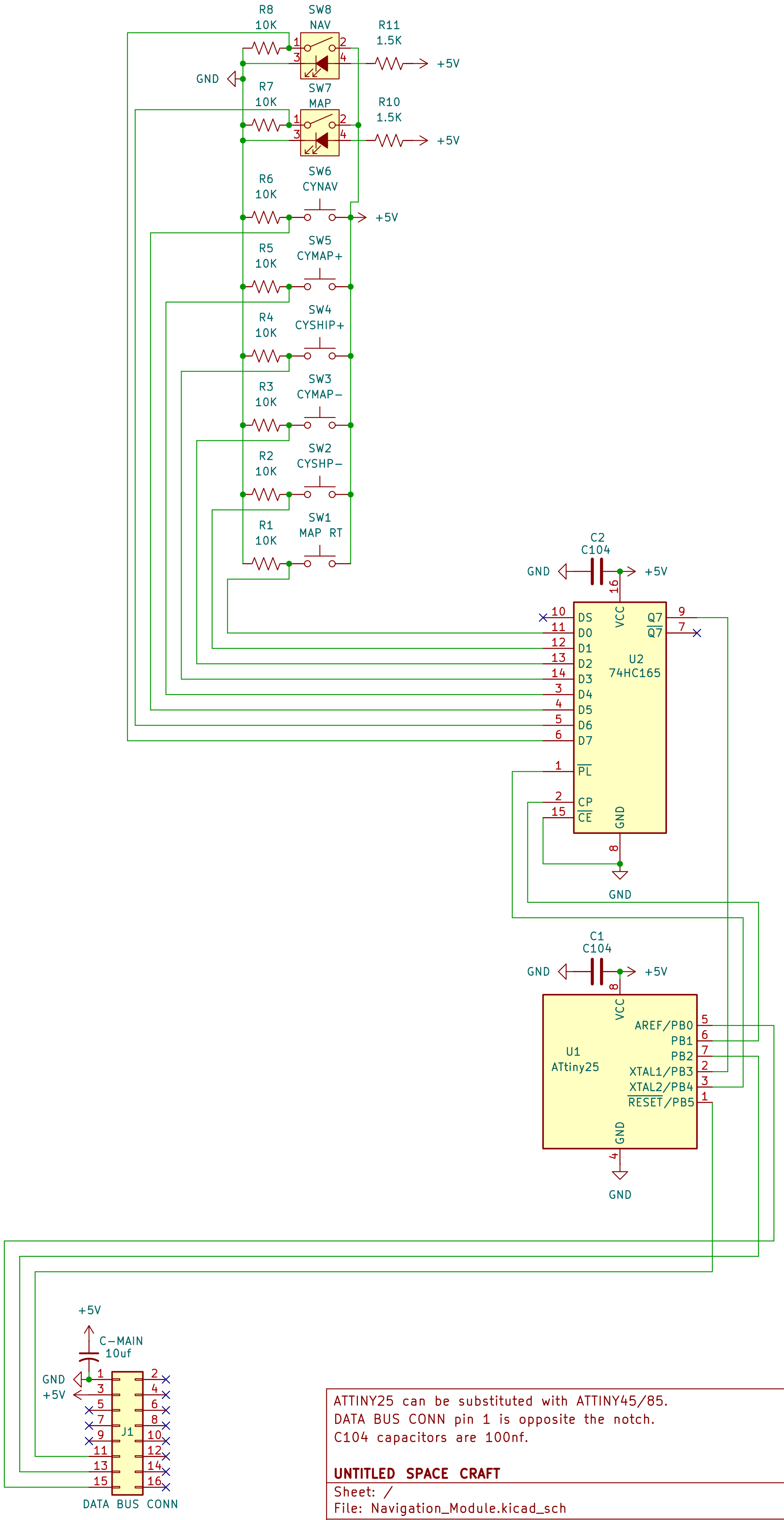
ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.
62k resistors used for blue LEDS, 10k for orange.

UNTITLED SPACE CRAFT

Sheet: /
File: Executive_Actions_G_C_Module.kicad_sch

Title: EXECUTIVE ACTIONS (GROUPS/CONTROL) MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.6**
KiCad E.D.A. 10.0.3		Id: 1/1



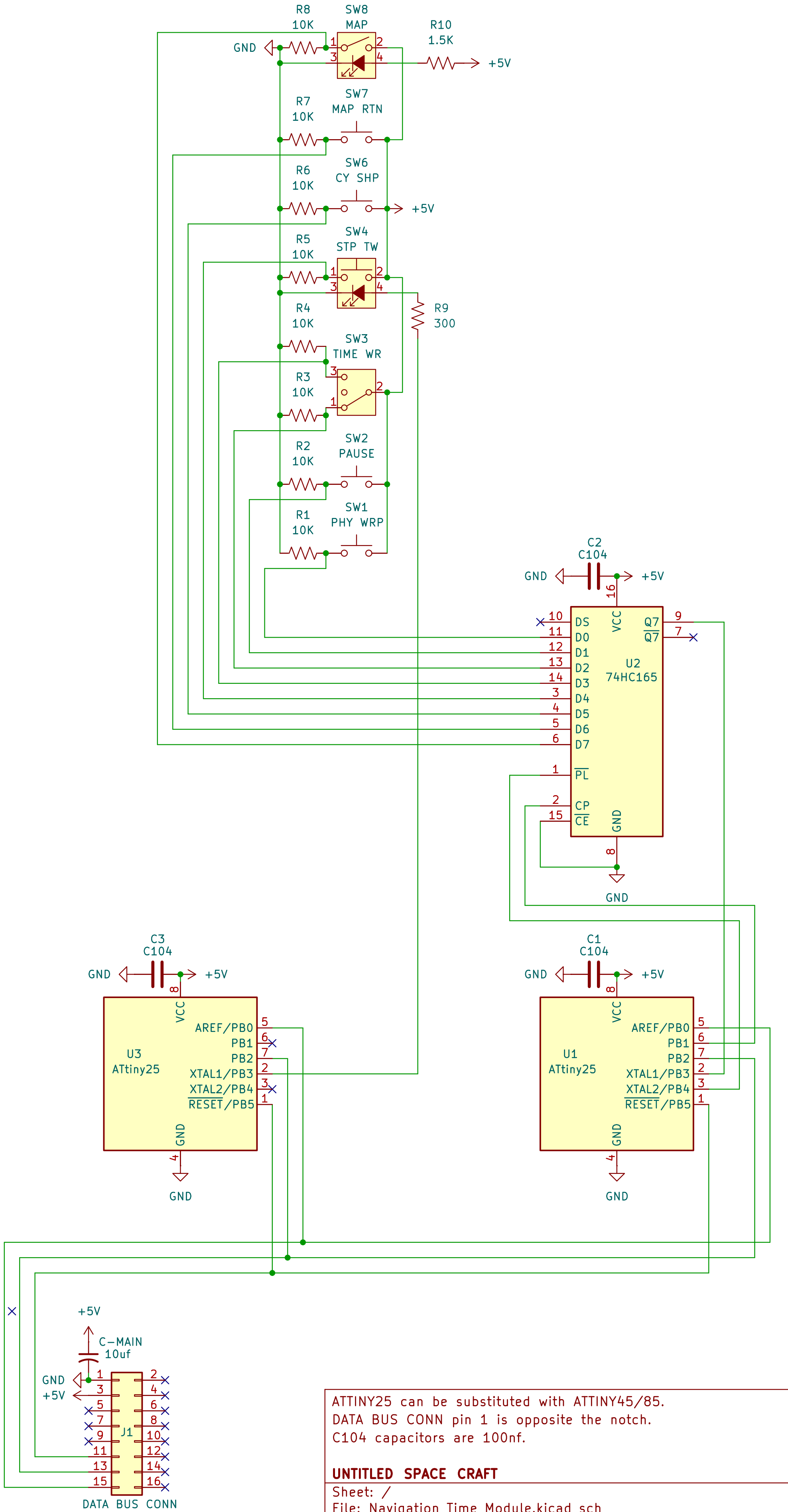
ATTINY25 can be substituted with ATTINY45/85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Navigation_Module.kicad_sch

Title: NAVIGATION MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



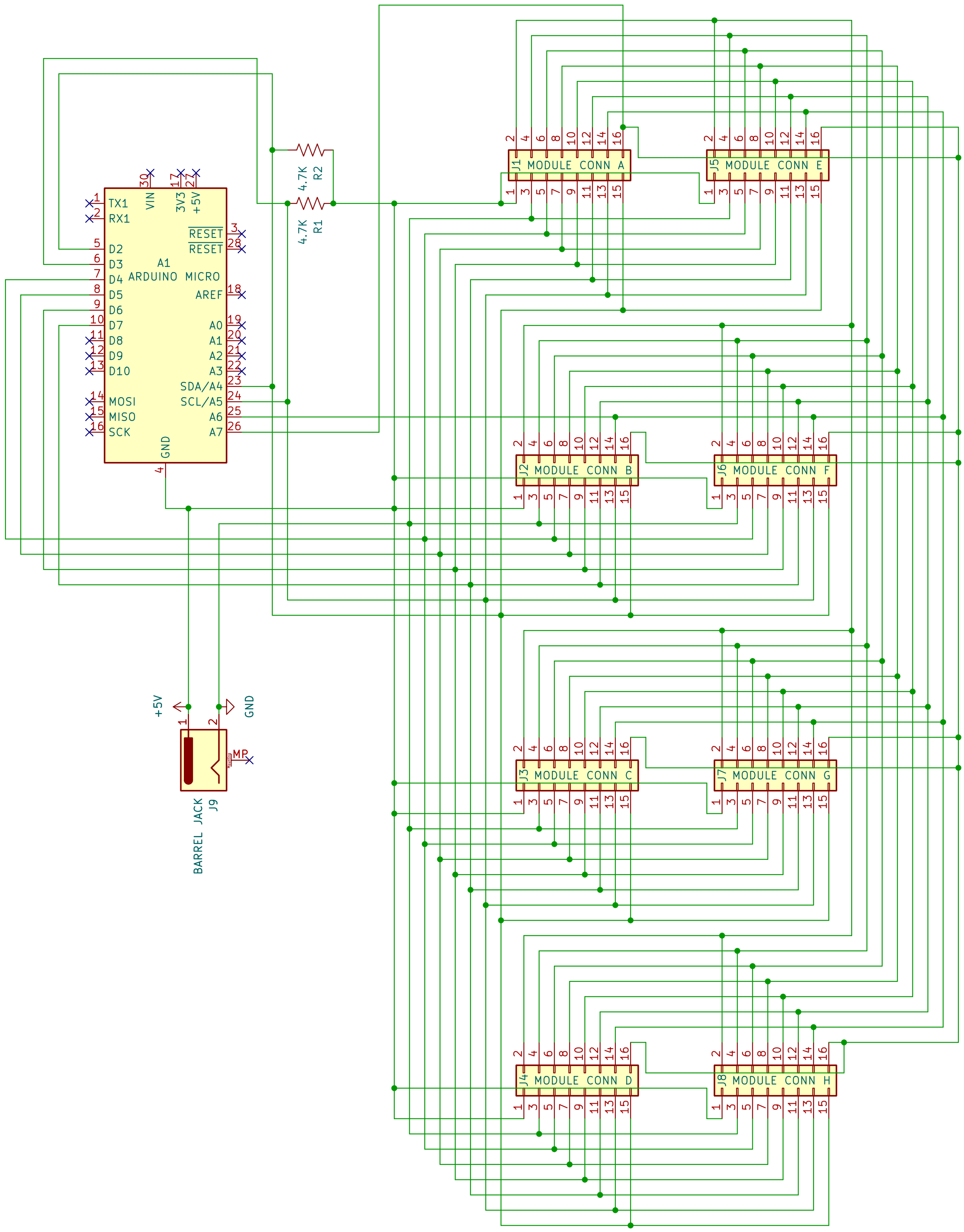
ATTINY25 can be substituted with ATTINY45/85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Navigation_Time_Module.kicad_sch

Title: NAVIGATION TIME MODULE

Size: A4 Date: 2026-06-17 **Rev: 1.7**
KiCad E.D.A. 10.0.3 Id: 1/1



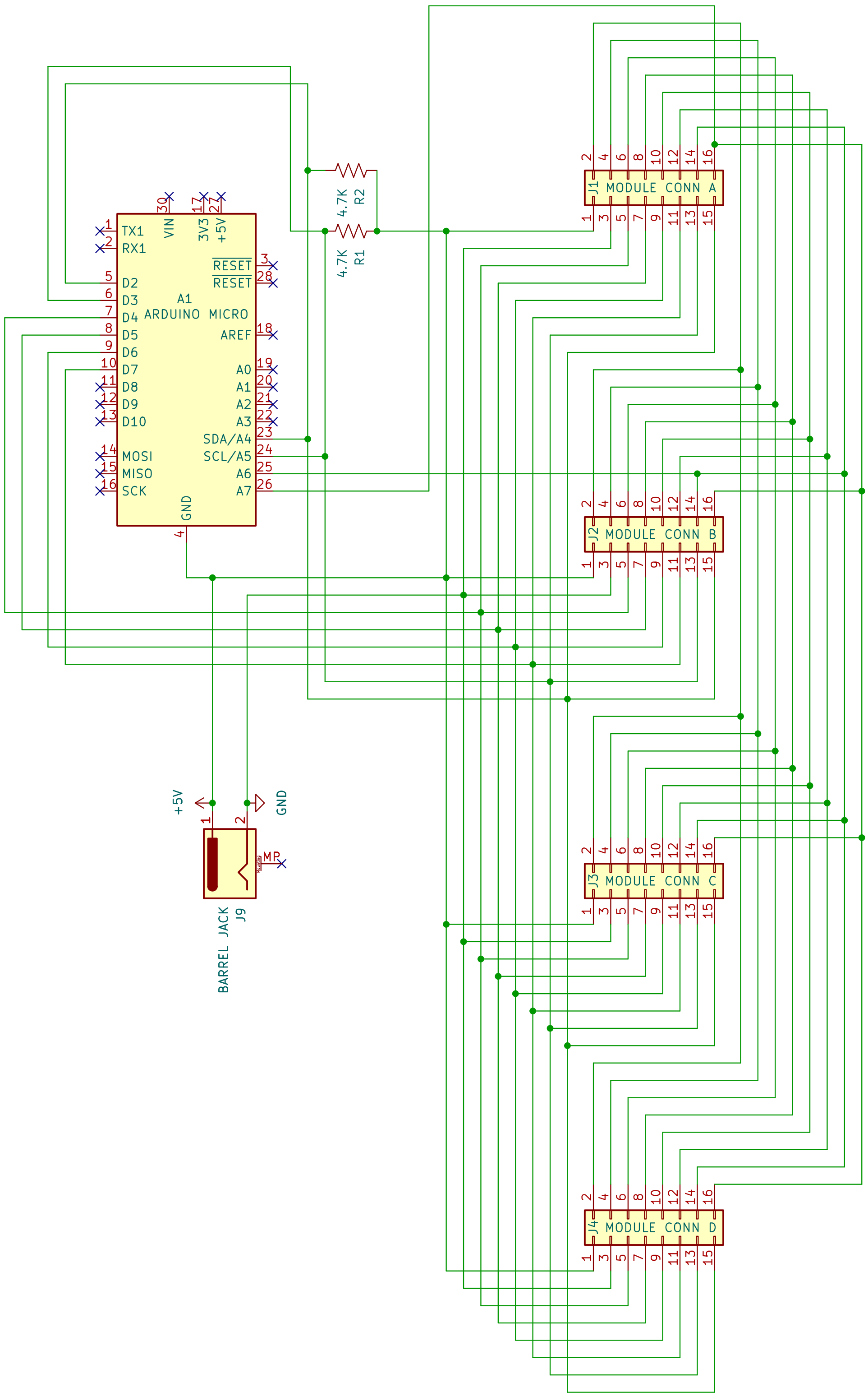
Pins are for Arduino Nano, but an Arduino Micro should fit into the same slot to enable keyboard emulation.
Barrel Jack: Tip/inside is 5V, Sleeve is GND.

UNTITLED SPACE CRAFT

Sheet: /
File: Octo_Hub.kicad_sch

Title: OCTO HUB

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



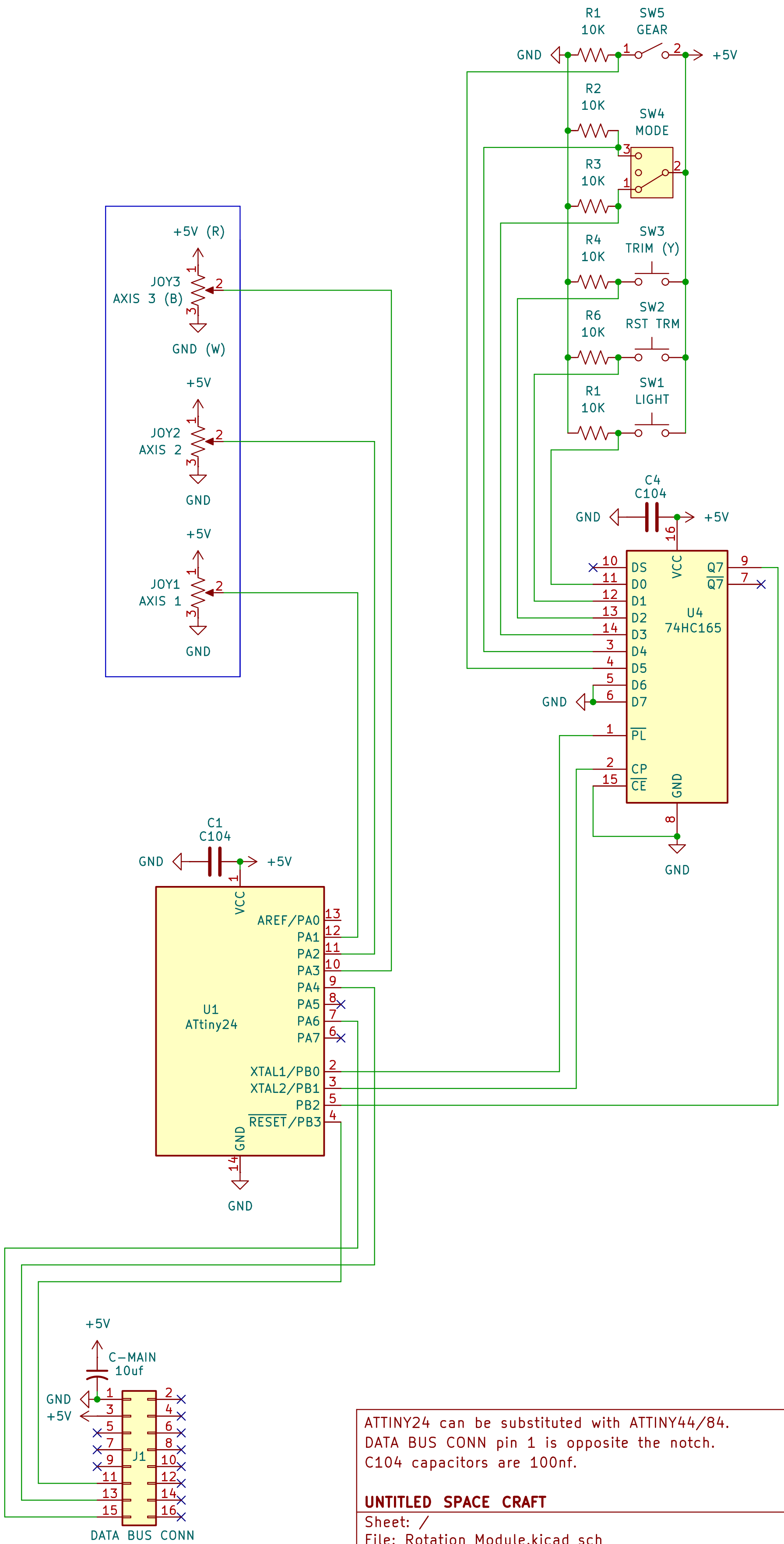
Pins are for Arduino Nano, but an Arduino Micro should
fit into the same slot to enable keyboard emulation.
Barrel Jack: Tip/inside is 5V, Sleeve is GND.

UNTITLED SPACE CRAFT

Sheet: /
File: Quad_Hub.kicad_sch

Title: QUAD HUB

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



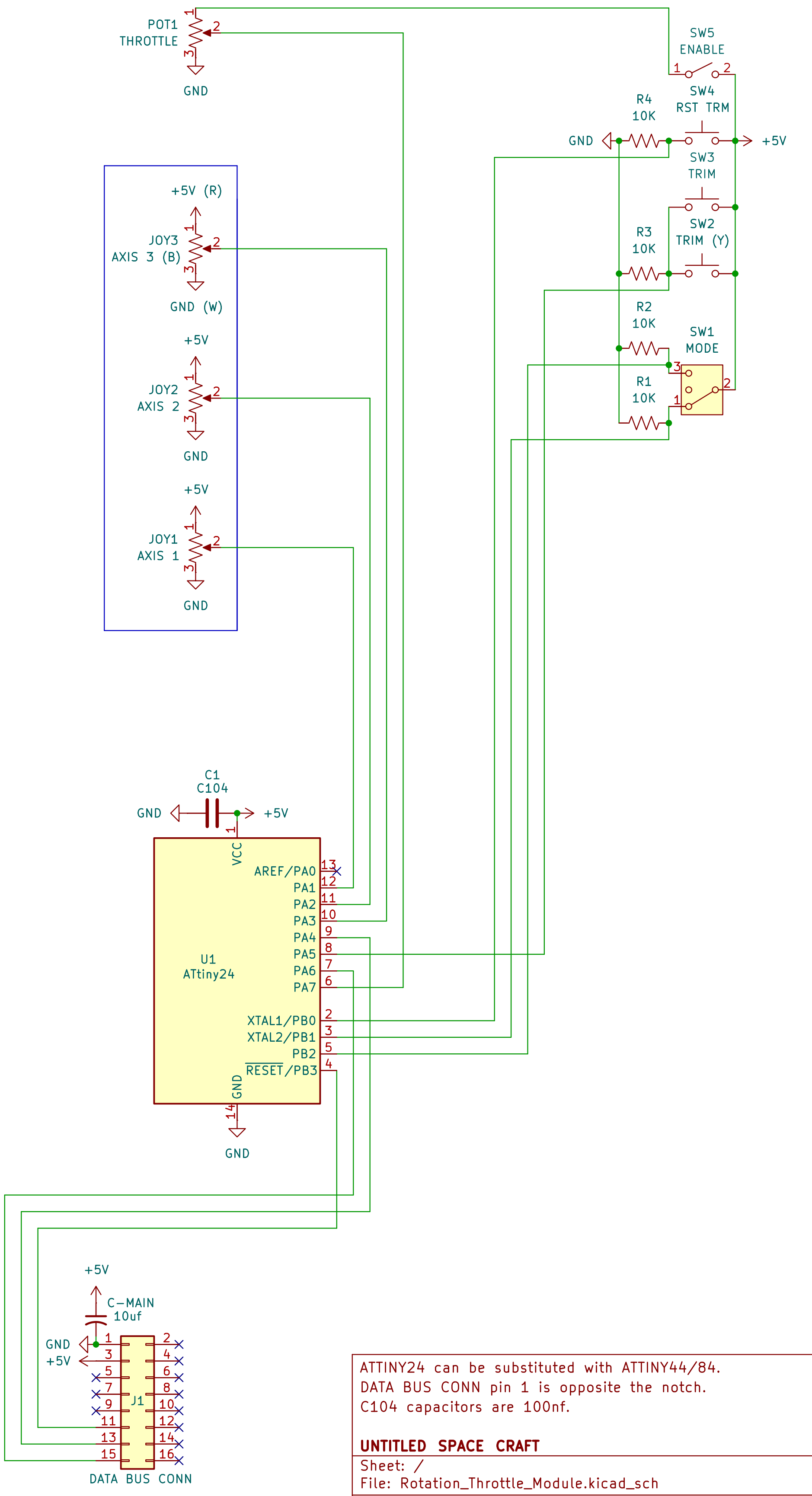
ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Rotation_Module.kicad_sch

Title: ROTATION MODULE

Size: A4 | Date: 2026-06-17 | **Rev: 1.7**
KiCad E.D.A. 10.0.3 | Id: 1/1



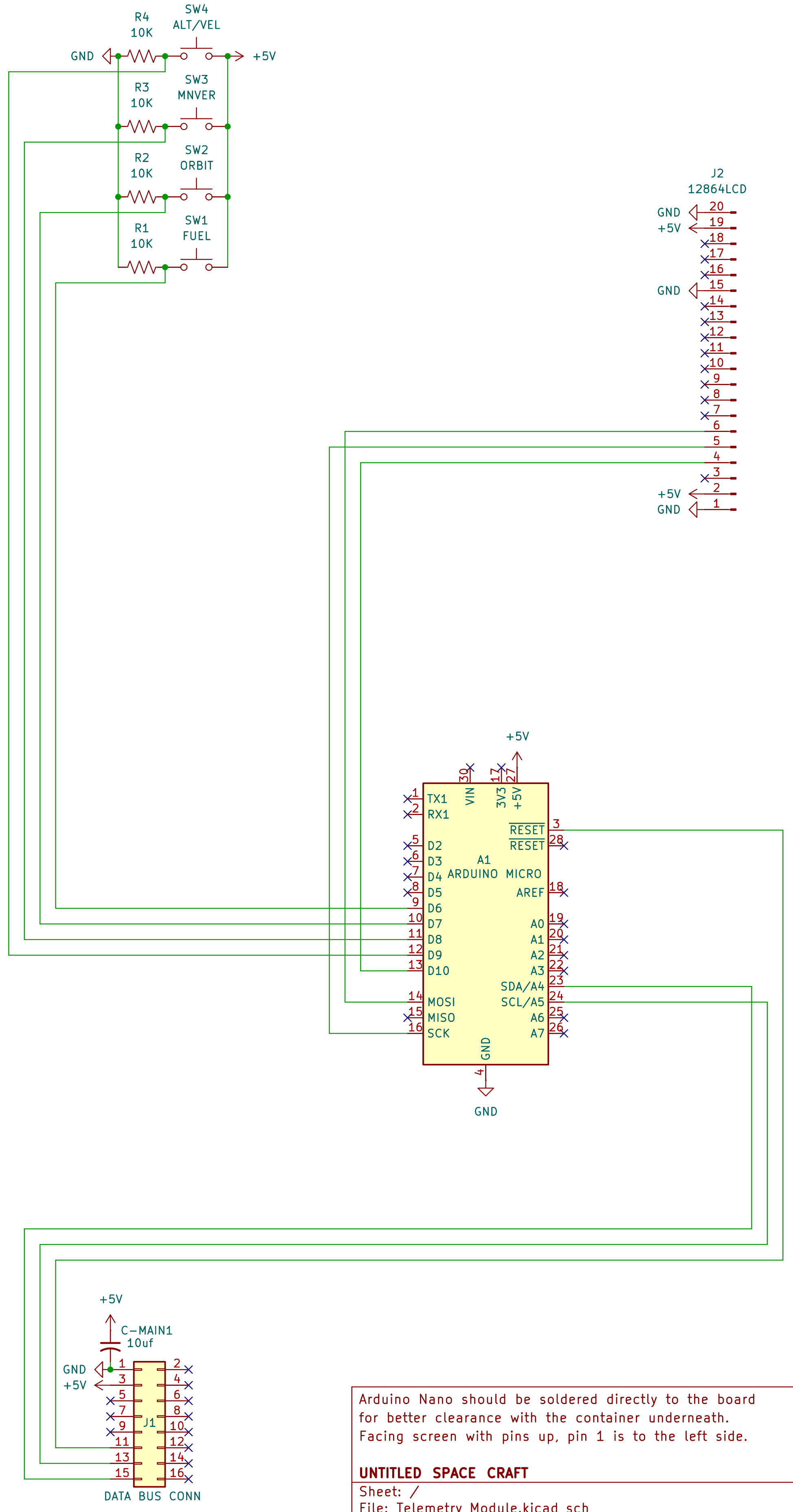
ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Rotation_Throttle_Module.kicad_sch

Title: ANALOG THROTTLE MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



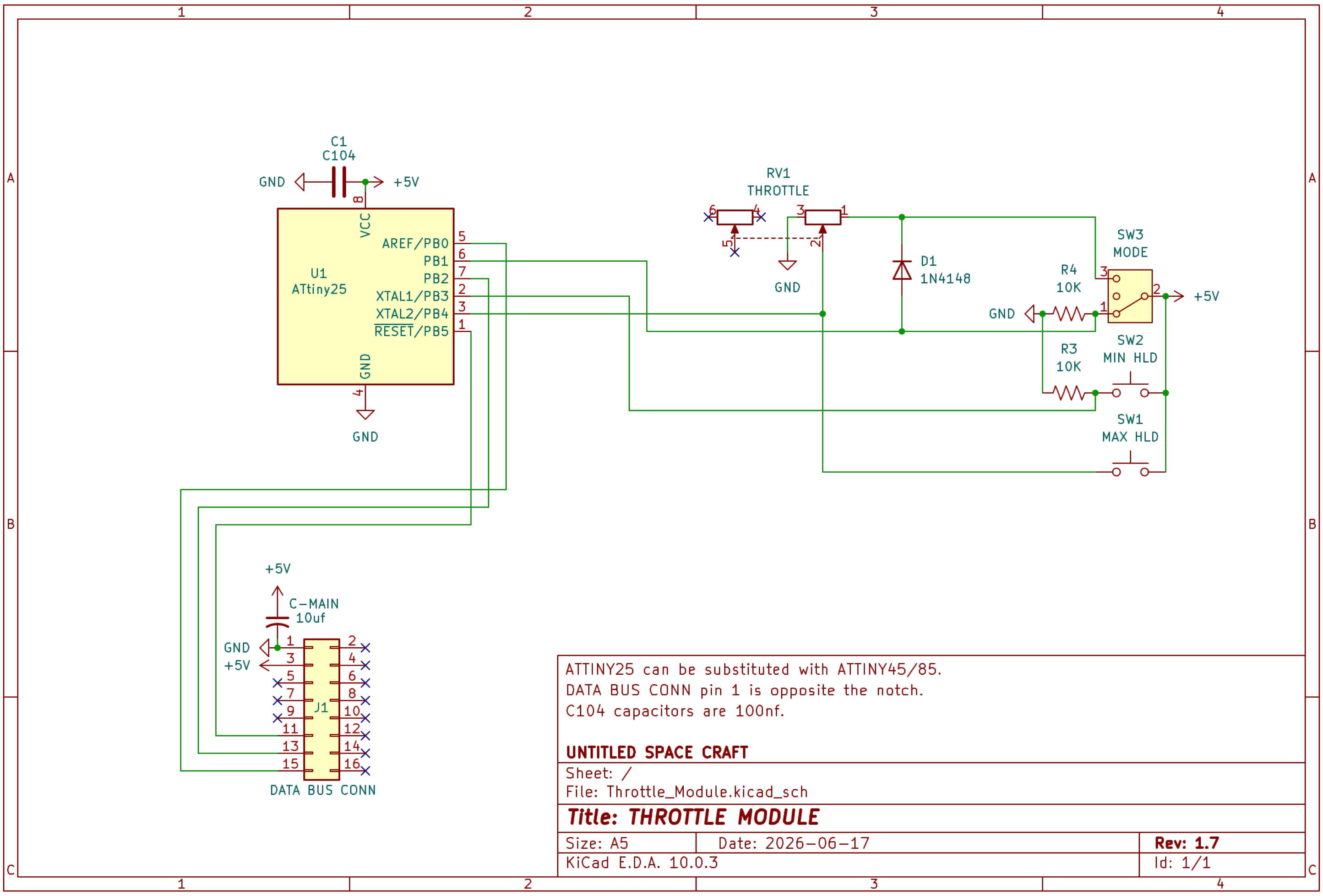
Arduino Nano should be soldered directly to the board
for better clearance with the container underneath.
Facing screen with pins up, pin 1 is to the left side.

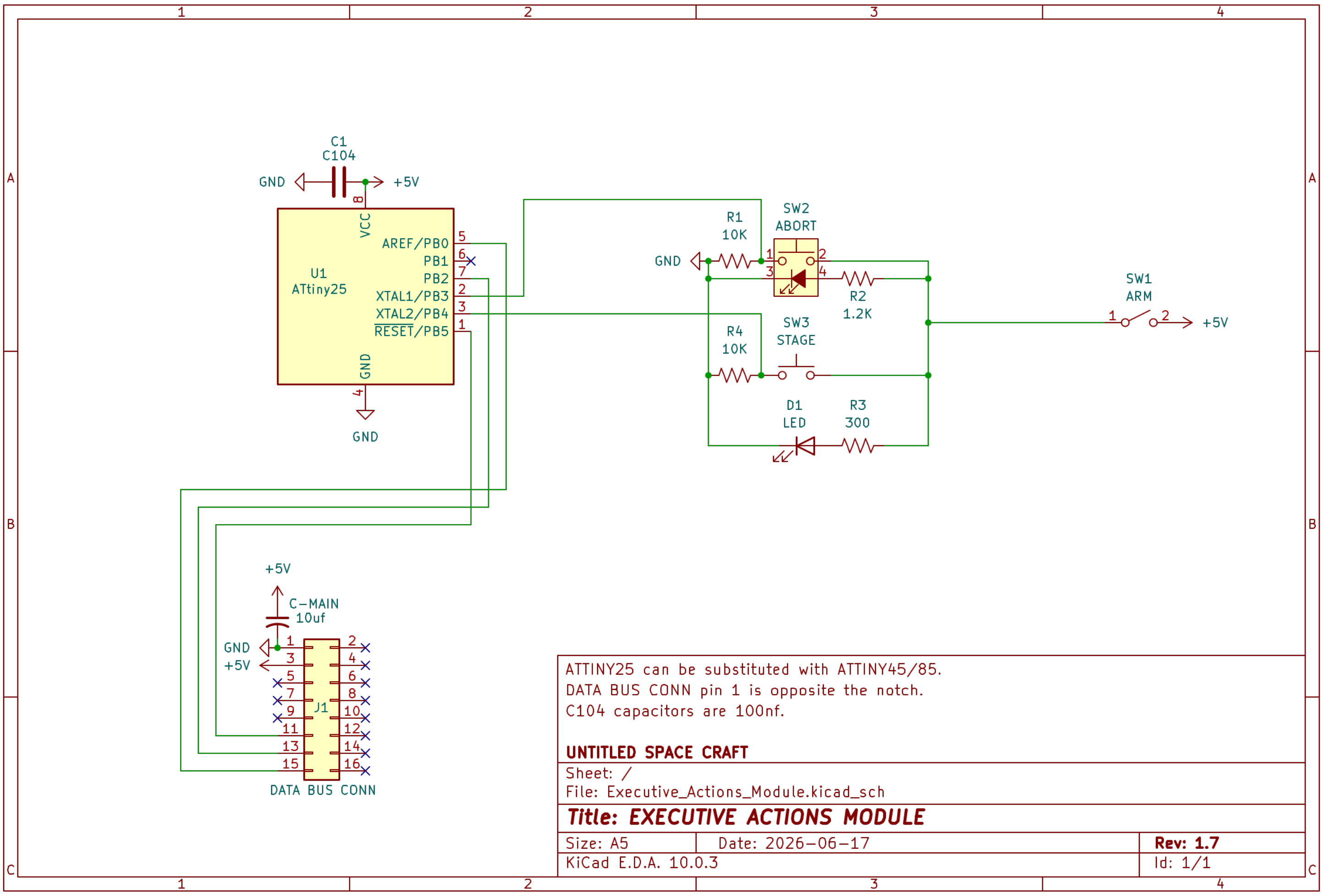
UNTITLED SPACE CRAFT

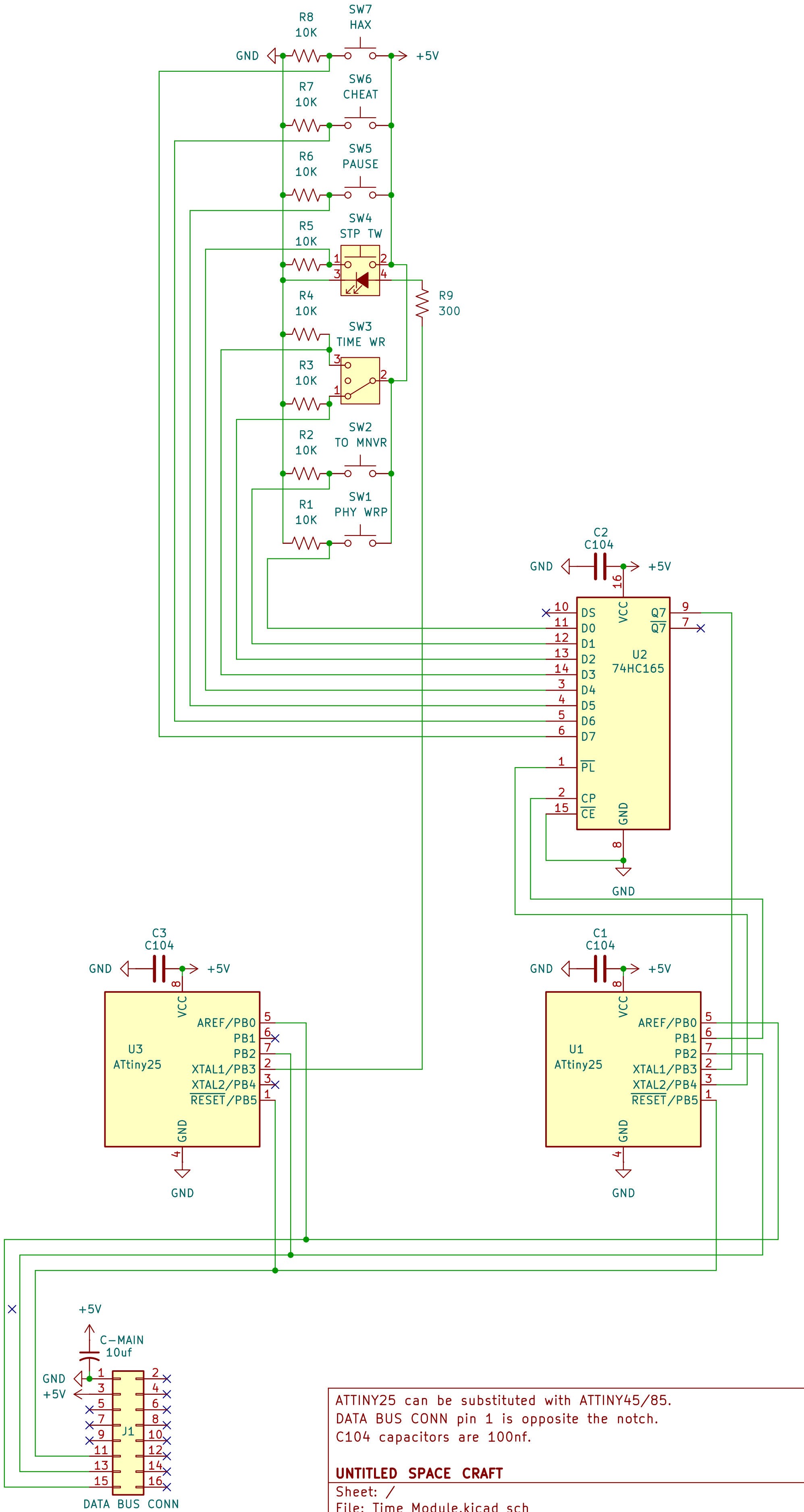
Sheet: /
File: Telemetry_Module.kicad_sch

Title: TELEMETRY MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1


C1
C104
GND
+5V
U1
ATtiny25
VCC
AREF/PB0
PB1
PB2
XTAL1/PB3
XTAL2/PB4
RESET/PB5
GND
RV1
THROTTLE
D1
1N4148
SW3
MODE
R4
10K
R3
10K
SW2
MIN HLD
SW1
MAX HLD
C-MAIN
10uf
J1
DATA BUS CONN
ATTINY25 can be substituted with ATTINY45/85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.
UNTITLED SPACE CRAFT
Sheet: /
File: Throttle_Module.kicad_sch
Title: THROTTLE MODULE
Size: A5
Date: 2026-06-17
Rev: 1.7
KiCad E.D.A. 10.0.3
Id: 1/1


C1
C104
GND
+5V
U1
ATtiny25
VCC
AREF/PB0
PB1
PB2
XTAL1/PB3
XTAL2/PB4
RESET/PB5
GND
R1
10K
SW2
ABORT
R2
1.2K
SW1
ARM
R4
10K
SW3
STAGE
D1
LED
R3
300
C-MAIN
10uf
J1
DATA BUS CONN
ATTINY25 can be substituted with ATTINY45/85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.
UNTITLED SPACE CRAFT
Sheet: /
File: Executive_Actions_Module.kicad_sch
Title: EXECUTIVE ACTIONS MODULE
Size: A5
Date: 2026-06-17
Rev: 1.7
KiCad E.D.A. 10.0.3
Id: 1/1



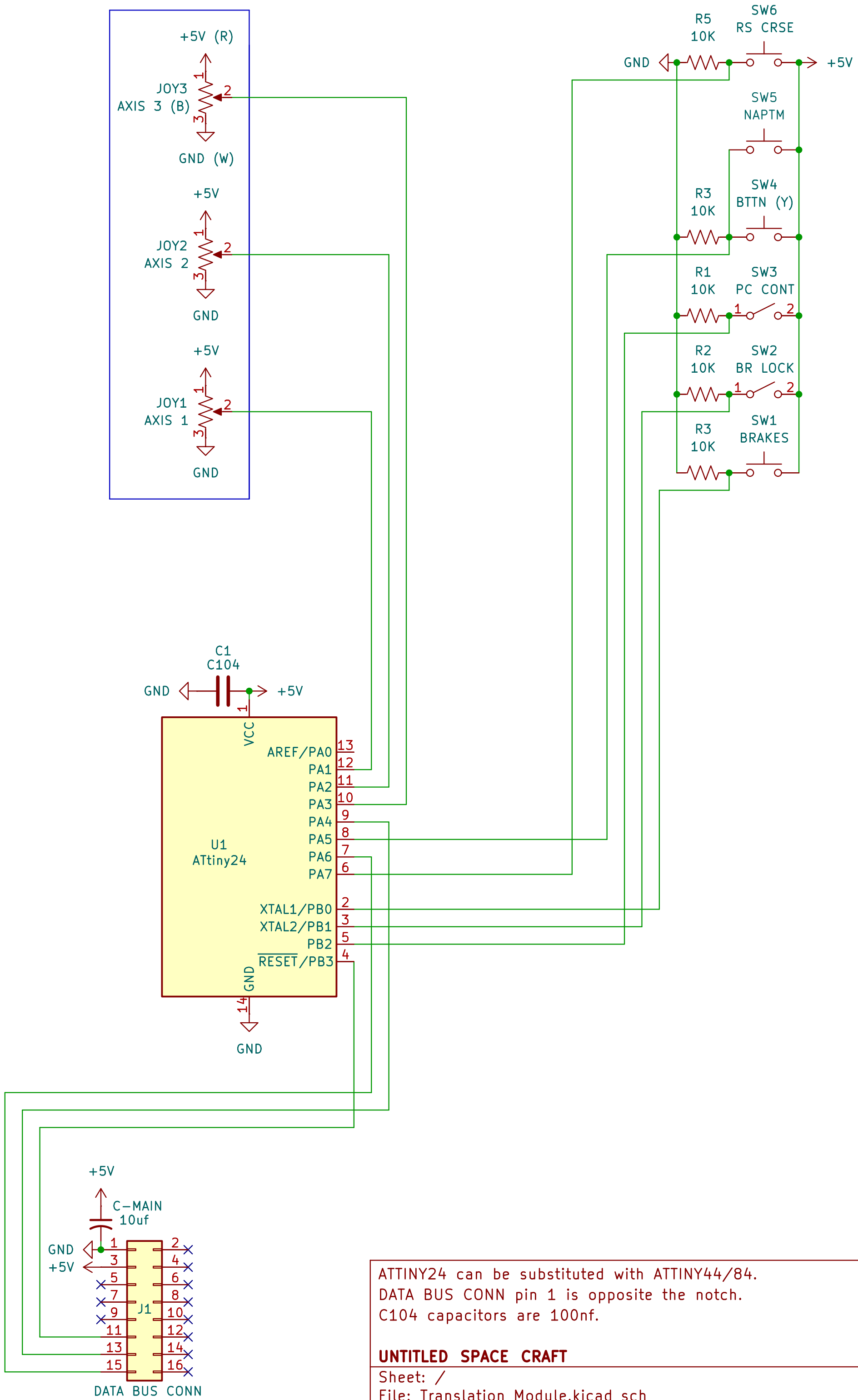
ATTINY25 can be substituted with ATTINY45/85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Time_Module.kicad_sch

Title: TIME MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



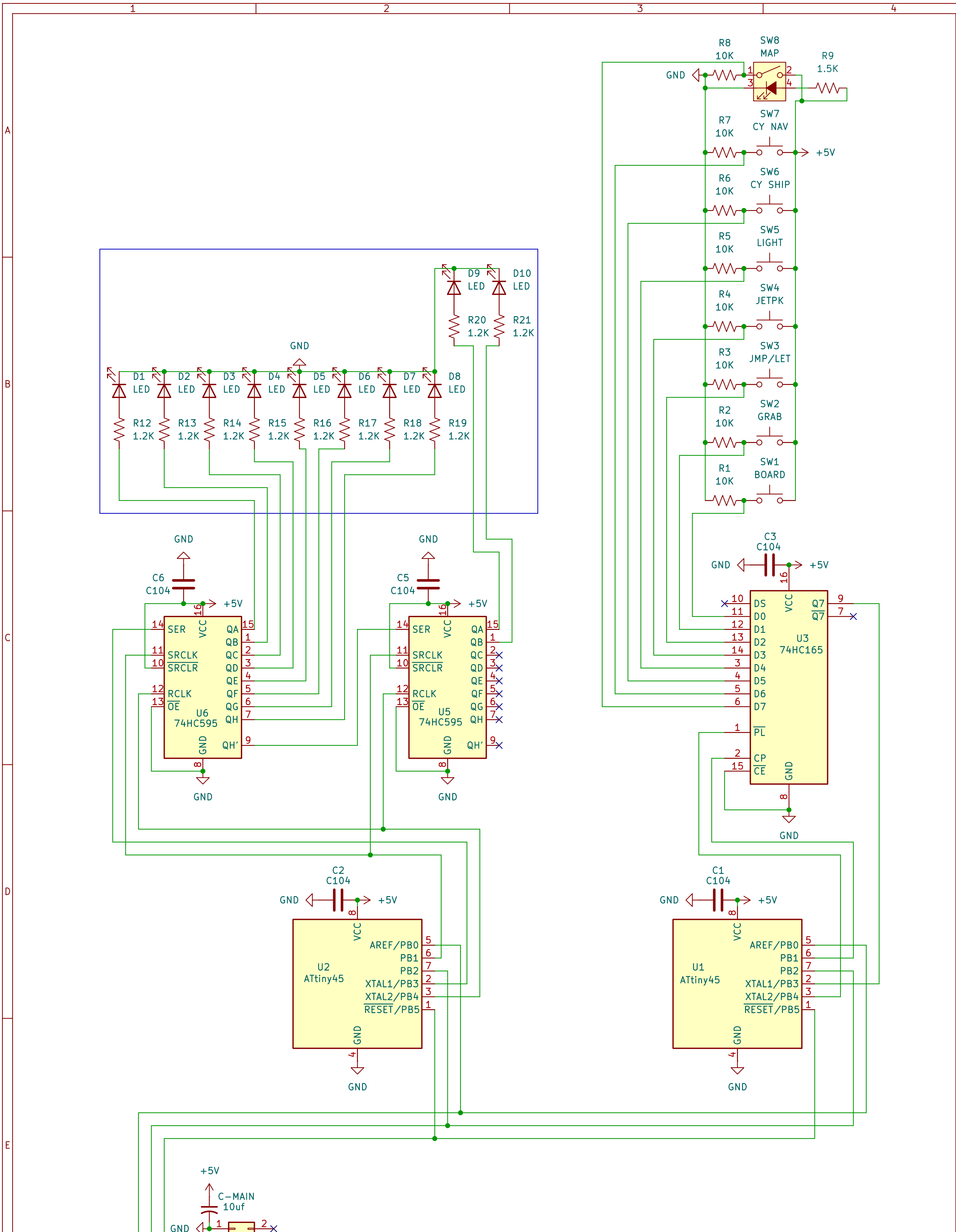
ATTINY24 can be substituted with ATTINY44/84.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Translation_Module.kicad_sch

Title: TRANSLATION MODULE

Size: A4	Date: 2026-06-17	**Rev: 1.7**
KiCad E.D.A. 10.0.3		Id: 1/1



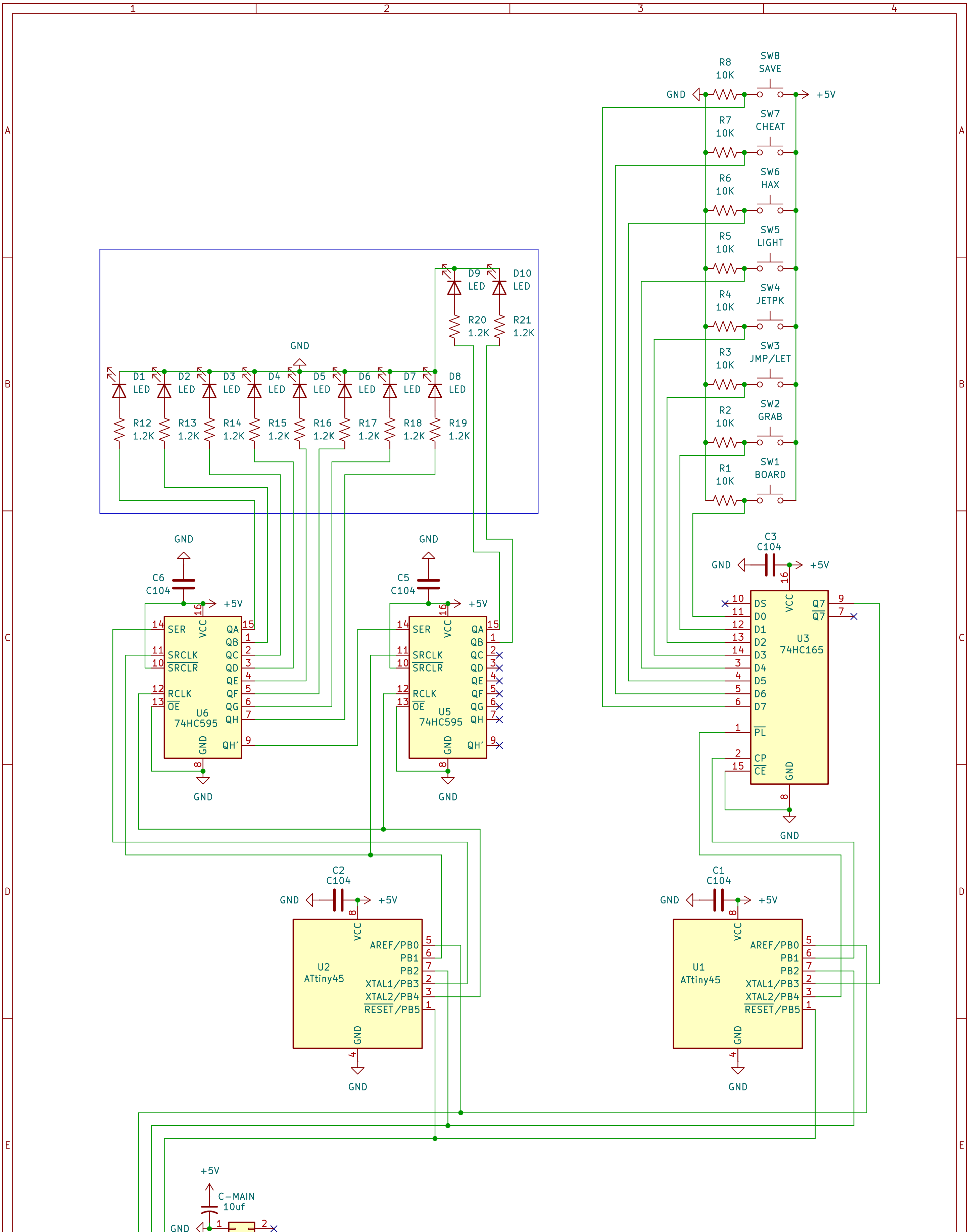
ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Utility_Navigation_Module.kicad_sch

Title: UTILITY NAVIGATION MODULE

Size: A4 | Date: 2026-06-17 | **Rev: 1.7**
KiCad E.D.A. 10.0.3 | Id: 1/1



ATTINY45 can be substituted with ATTINY85.
DATA BUS CONN pin 1 is opposite the notch.
C104 capacitors are 100nf.

UNTITLED SPACE CRAFT

Sheet: /
File: Utility_Time_Module.kicad_sch

Title: UTILITY TIME MODULE

Size: A4	Date: 2026-06-17
KiCad E.D.A. 10.0.3	
Rev: 1.7	Id: 1/1